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Optimal Currency Bands

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Abstract

We provide an analytical solution to the steady-state optimal currency-band problem, which prior target-zone literature addressed only numerically. A monetary authority chooses a band to trade off the steady-state variance of the exchange rate against proportional intervention costs. Building on [Krugman \(1991\)](#), we derive an exact closed-form expression for the steady-state variance of the exchange rate, which is approximately proportional to the squared band width. We solve the authority's problem analytically, obtaining closed-form optimal bands in the narrow- and wide-band limits, and show that the optimal band depends only on the ratio of intervention cost to fundamental volatility. Calibrating the model to daily Italian lira data from the ERM I period – which spans the lira's 1990 transition from a $\pm 6\%$ to a $\pm 2.25\%$ band – we recover structural estimates of fundamental volatility and intervention cost for each regime from the model's exact equilibrium conditions, since the calibrated ratio of cost to volatility lies between the two closed-form approximations' regions of validity. The estimated intervention costs are roughly two to fifteen times larger than in comparable numerical exercises ([Miller and Zhang, 1996](#)). We argue this gap partly reflects a specification bias rather than a purely structural difference: the model assumes a symmetric intervention cost across both boundaries, but the lira faced near-continuous one-sided pressure against the depreciation boundary, so fitting a single symmetric cost parameter to data generated almost entirely by defense of one (likely costlier) boundary mechanically inflates the estimate and plausibly biases the model's implied bands upward relative to a specification with boundary-specific costs and explicit realignment risk. A cross-regime counterfactual shows that both fundamental volatility and intervention cost contribute to the observed change in band width, with the volatility channel accounting for the larger share; yet neither channel alone, nor both together, rationalizes the observed $\pm 2.25\%$ width, suggesting it served purposes beyond cost minimization such as commitment value and a precautionary buffer against speculative attacks. The model abstracts from parity-realignment risk and assumes symmetric intervention costs across boundaries – the two main caveats to the empirical application.

Keywords: Krugman target zones, exchange rate mechanism, regulated Brownian motion, Italian lira

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Replication code (MATLAB) and the latest version of this thesis are available at: https://github.com/ianasekerina/optimal_currency_bands.

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1 Introduction

How wide should an exchange-rate target zone be? Under a target-zone regime, the monetary authority commits to keeping the exchange rate within predetermined upper and lower bounds around a central parity. Such arrangements occupy an intermediate position between fixed and freely floating exchange rates: the exchange rate is allowed to respond to market forces within the band, while policy intervention limits excessive deviations from the target. Target zones have played an important role in international monetary history, one example is the European Monetary System (EMS), where participating currencies were allowed to fluctuate within agreed margins around central parities. The width of these margins varied across countries and over time, reflecting different degrees of exchange-rate flexibility and policy commitment.

The theory of target zones originates with [Krugman \(1991\)](#), who shows how the exchange rate evolves within a predetermined band when market participants anticipate future intervention at the boundaries. His framework generates the honeymoon effect, whereby expectations of future intervention stabilize the exchange rate even before the boundaries are reached. While this literature provides a rich characterization of exchange-rate dynamics within a given target zone, the width of the band itself is typically taken as exogenous.

This observation raises a natural policy question. If intervention is costly, narrow bands may require frequent policy actions to defend the target zone. Conversely, wider bands reduce intervention frequency but allow greater exchange-rate volatility. How should a policymaker balance these competing objectives when choosing the width of the band? This work addresses that question within Krugman’s target-zone framework. We formulate a steady-state planner’s problem that trades off exchange-rate volatility against intervention costs and characterize the optimal exchange-rate band.

The work makes three contributions. First, it derives an exact closed-form expression for the steady-state variance of the exchange rate under a target zone. Second, it shows that this expression admits two economically meaningful asymptotic approximations corresponding to narrow- and wide-band regimes. These approximations yield closed-form solutions for the planner’s optimal band and allow transparent comparative statics linking optimal policy to intervention costs and fundamental volatility. Third, the model is evaluated using data from Italy’s participation in the Exchange Rate Mechanism (ERM I), exploiting the transition from a $\pm 6\%$ to a $\pm 2.25\%$ fluctuation band in 1990.

Calibrating the model to daily lira data from the ERM I period yields two main findings. First, the implied intervention-cost parameter is substantially larger than values typically assumed in numerical exercises, reflecting the persistent pressure faced by the lira near the depreciation boundary. Second, the counterfactual exercise decomposes the observed change in band width into two structural channels: changes in fundamental volatility and changes in intervention costs. Recombining parameter estimates across regimes shows that each channel accounts for part of the observed narrowing of the band, while neither is sufficient on its own. The transition from the $\pm 6\%$ to the $\pm 2.25\%$ regime therefore reflects the joint influence of changing macroeconomic conditions and changing intervention incentives.

Related Literature The target zone literature originates with [Krugman \(1991\)](#), who demonstrated that credible commitment to a band stabilizes the exchange rate through the honeymoon effect. Subsequent empirical work documented that exchange rates spend disproportionate time near the boundaries rather than the interior, a pattern [Bertola and Caballero \(1992\)](#) attributed to anticipation of discrete parity realignments, which they formalize in a model with state-dependent realignment probabilities. [Svensson \(1991\)](#) shows that within narrow bands the interest rate differential’s conditional volatility is high and increasing as the zone narrows, while [Svensson \(1994\)](#) argues that finite bands grant central banks meaningful monetary independence even under free capital mobility. [Svensson \(1992\)](#) surveys the early literature and explicitly notes that the question of optimal bandwidth had not been resolved.

The closest antecedents in the optimal band literature are [Miller and Zhang \(1996\)](#) and [Im \(2001\)](#). [Miller and Zhang](#) frame the problem as a credibility trade-off: discretionary bands are wider and less stabilizing than bands set under precommitment, and they argue this provides a theoretical rationale for an exchange rate mechanism. [Im](#) solves explicitly for optimal bands in an infinite-horizon framework with interest rate deviation costs and derives comparative statics. Both papers rely on numerical simulations rather than closed-form solutions and do not undertake structural estimation. The present work contributes exact variance and optimal band formulae, a homogeneity result that reduces the parameter space to a single ratio κ/σ , and a calibration to Italian ERM I data that disciplines the model with observable moments.

The analytical structure of the planner’s problem is related to the cash-management literature initiated by [Miller and Orr \(1966\)](#) and [Constantinides and Richard \(1978\)](#), who characterize (s, S) policies for inventory and cash balances under proportional adjustment costs. The exchange rate band problem shares the same mathematical structure, regulated Brownian motion with proportional intervention costs, but the objective differs: here the authority minimizes exchange rate variance rather than holding or shortage costs. Finally, [Alvarez \(2025\)](#) extends the target zone framework to time-varying bands showing that a pre-announced widening path can enhance flexibility but may entail significant transitional intervention costs; that work provides the institutional motivation for the Argentine 2024 episode.

2 Target-Zone Environment

2.1 Exchange Rate Dynamics

The Exchange Rate Model The model is that of a small open economy built on three standard equilibrium relationships. Let s_t denote the log nominal exchange rate, p_t and p_t^* the logs of domestic and foreign price levels, i_t and i_t^* the domestic and foreign nominal interest rates, m_t the log money supply, and v_t an exogenous velocity shock. Purchasing Power Parity (PPP),

$$p_t = s_t + p_t^*, \quad (1)$$

links the domestic price level to the foreign price level expressed in domestic currency. Uncovered Interest Parity (UIP),

$$i_t = i_t^* + \mathbb{E}_t[ds_t/dt], \quad (2)$$

ties the interest differential to expected depreciation. Finally, equilibrium in the money market requires

$$m_t - p_t + v_t = -\alpha i_t, \quad (3)$$

where $\alpha > 0$ is the semi-elasticity of money demand with respect to the interest rate.

PPP states that the domestic price level equals the foreign price level expressed in domestic currency. It provides a long-run anchor for the exchange rate: higher domestic prices relative to abroad imply a depreciation. Equation UIP links interest rate differentials to expected exchange rate movements. If domestic interest rates are higher than foreign rates, investors must expect a depreciation of the domestic currency. Real money balances depend negatively on the nominal interest rate. The term v_t captures exogenous velocity (or money demand) shocks.

Substituting (1) into (3) and using (2) to eliminate i_t , we obtain

$$s_t = m_t + v_t - p_t^* + \alpha \left(i_t^* + \mathbb{E}_t \left[\frac{ds_t}{dt} \right] \right).$$

This expression highlights that the exchange rate depends on both current fundamentals and expectations of future depreciation.

Define the fundamental as

$$x_t := \frac{m_t + v_t - p_t^*}{\alpha} + i_t^*.$$

The fundamental x_t summarizes monetary and external forces driving the exchange rate: higher domestic money supply or velocity raises x_t , while higher foreign interest rates lower it. Then the exchange rate satisfies

$$s_t = \alpha x_t + \alpha \mathbb{E}_t \left[\frac{ds_t}{dt} \right].$$

Dividing by α and defining $\beta := 1/\alpha$, we obtain

$$\beta s_t = x_t + \mathbb{E}_t[ds_t/dt] \quad (4)$$

This is a standard forward-looking asset pricing equation: the exchange rate depends on current fundamentals and expected future changes.

The parameter α measures the sensitivity of money demand to interest rates. A higher α implies that expectations of future depreciation have a stronger impact on current equilibrium.

Finally, we assume that the fundamental follows a Brownian motion without a drift

$$dx_t = \sigma dW_t,$$

where σ represents volatility.

Exchange Rate Under Symmetric Fixed Bands With the model in hand, we now characterize the equilibrium exchange rate when the central bank commits to keeping s_t within a symmetric band $[-b, b]$. The key insight, due to [Krugman \(1991\)](#), is that the credible commitment itself stabilizes the exchange rate before the boundaries are reached. The following proposition formalizes and extends the results of [Krugman \(1991\)](#) and [Alvarez \(2025\)](#).

Proposition 1. *Suppose that the fundamental has zero drift, i.e. $dx_t = \sigma dW_t$, then the following results hold*

1. *Under a fixed currency band, the exchange rate s as a function of the fundamentals x is given by*

$$s = f(x) = \frac{1}{\beta} \left(x - \frac{1}{\eta} \frac{\sinh(\eta x)}{\cosh(\eta X)} \right), \quad x \in [-X, X] \quad \eta = \sqrt{\frac{2\beta}{\sigma^2}}; \quad (5)$$

2. *The boundary X solves*

$$b = \frac{1}{\eta\beta} (\eta X - \tanh(\eta X)); \quad (6)$$

3. *The boundary X is increasing in b , and*

$$\frac{\partial X}{\partial b} = \left(\frac{\partial b}{\partial X} \right)^{-1} = \frac{\beta}{1 - \text{sech}^2(\eta X)} > 0 \quad (7)$$

The mapping f is S-shaped and always lies strictly inside the free-float line $s = x/\beta$ (as illustrated in [Figure 1](#)). Intuitively, when fundamentals approach the boundary, market participants expect intervention, which moderates the current exchange rate before any intervention actually takes place. This is known as the honeymoon effect.

Additionally, the parameter $\eta = \sqrt{2\beta/\sigma^2}$ controls the curvature of f . When η is large (low volatility or high β), the S-curve is nearly flat, so the band is very effective at insulating the exchange rate from fundamentals. When η is small (high volatility), the curve is steep and the exchange rate tracks fundamentals more closely.

The result (7) establishes that X and b are interchangeable as choice variables for the central bank: optimizing over the exchange-rate band b is equivalent to optimizing over the fundamental boundary X , sometimes referred to as the intervention trigger. We exploit this equivalence throughout.

The smooth-pasting conditions $f'(X) = f'(-X) = 0$ reflect the fact that intervention occurs only at the boundaries, and the marginal cost of the exchange rate reaching the band edge must be zero (no discrete jumps in the rate of intervention). This is the key economic optimality condition of the model.

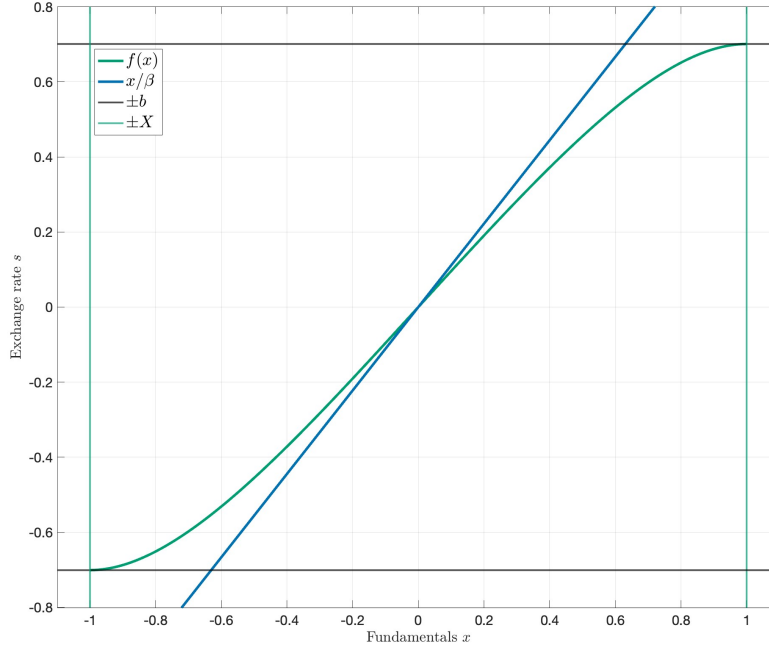


Figure 1: Exchange rate $f(x)$ under a target zone (teal) versus a free float $s = x/\beta$ (blue). The S-shaped curve lies strictly inside the free-float line.

Steady-State Distribution and Variance To evaluate the authority's loss function, we need the long-run distribution of the fundamental and the resulting variance of the exchange rate. We proceed in two steps: first, derive the stationary density of x_t under reflecting barriers; second, integrate the squared Krugman function against that density.

Lemma 1. *The steady-state distribution for the fundamental following Brownian motion with reflecting barriers is*

$$m(x; X) = \frac{1}{2X}, \quad x \in [-X, X] \quad (8)$$

The uniform distribution has a transparent interpretation: with no drift, the fundamental spends equal time in every subinterval of $[-X, X]$, so the long-run density is flat.

The steady-state variance is defined as

$$\text{Var}(s) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^T s_t^2 dt.$$

By the Birkhoff-Khinchin ergodic theorem, the steady-state variance will be equal to the integral of $f^2(x)$ under the invariant density (8). This simplifies the variance calculation:

$$\text{Var}(s) = \frac{1}{2X} \int_{-X}^X f^2(x) dx. \quad (9)$$

Proposition 2 (Exact Variance). *If the exchange rate follows*

$$s = f(x) = \frac{1}{\beta} \left(x - \frac{1}{\eta} \frac{\sinh(\eta x)}{\cosh(\eta X)} \right), \quad x \in [-X, X], \quad \eta = \sqrt{\frac{2\beta}{\sigma^2}},$$

then its stationary (steady-state) variance is given by

$$\text{Var}(s) = \frac{1}{2X\beta^2} \left(\frac{2X^3}{3} - \frac{4X}{\eta^2} + \frac{5 \tanh(\eta X)}{\eta^3} - \frac{X}{\eta^2 \cosh^2(\eta X)} \right) \quad (10)$$

The structure of (10) reflects the two forces shaping exchange rate volatility inside a band. The leading term $X^2/(3\beta^2)$ is what the variance would be if f were linear. Wider bands expose the exchange rate to more fundamental variation. The remaining terms, all involving $\tanh$ and $\cosh$, capture the honeymoon effect (Krugman, 1991): the S-shaped curvature of f compresses the exchange rate near the boundaries, pulling variance below what a linear model would predict.

Approximations The exact variance (10) is closed form but unwieldy for optimization. To obtain intuition and closed-form solutions, we derive two complementary approximations. One valid for narrow (X is small) bands, the other for wide (X is large) bands. These two regimes have distinct economic interpretations: narrow bands correspond to hard pegs where intervention is frequent, while wide bands approximate a managed float where intervention is rare.

Corollary 1 (Approximated Variance under Symmetric Bands). *Suppose that the exchange rate follows*

$$s = f(x) = \frac{1}{\beta} \left(x - \frac{1}{\eta} \frac{\sinh(\eta x)}{\cosh(\eta X)} \right), \quad x \in [-X, X], \quad \eta = \sqrt{\frac{2\beta}{\sigma^2}}.$$

Then the steady-state variance can be approximated as follows

1. *For X small:*

$$\text{Var}(s) \approx \frac{68}{315} \frac{X^6}{\sigma^4} \quad \text{or, equivalently,} \quad \text{Var}(s) \approx \frac{17}{35} b^2; \quad (11)$$

2. *For X large:*

$$\text{Var}(s) \approx \frac{X^2}{3\beta^2} \quad \text{or, equivalently,} \quad \text{Var}(s) \approx \frac{1}{3} b^2. \quad (12)$$

The constant $17/35$ exceeds $1/3$, implying that for a given band width b , the steady-state variance of the exchange rate is higher under the narrow-band approximation than under the wide-band one. This result reflects the differing curvature of the Krugman function across regimes. In

the wide-band limit, the S-curve is nearly flat and the exchange rate tracks fundamentals almost linearly; the variance is then simply $b^2/3$, the variance of a uniform distribution on $[-b, b]$. In the narrow-band limit, the S-curve is more sharply curved, and the exchange rate compresses less of the fundamental variation than the linear approximation would suggest, yielding a proportionality constant closer to one-half.

2.2 Planner's Problem

Problem formulation We consider a monetary authority that manages the exchange rate by adjusting its stock of foreign exchange (FX) reserves. When the authority intervenes in FX markets, its reserve position changes, and interventions occur only when fundamentals hit the boundaries. If $x = X^*$, then the FX reserves are sold, and if $x = -X^*$, then the FX reserves are bought. We define two increasing, continuous, and positive processes: dU_t – a purchasing process (upward adjustment of reserves, happens when $x = -X^*$), and dD_t – the selling process (downward adjustment of reserves, happens when $x = X^*$). Then the fundamental x_t evolves under central-bank interventions as:

$$x_t = x_0 + \sigma W_t + \int_0^t dU_s - \int_0^t dD_s$$

$$dx_t = \sigma dW_t + dU_t - dD_t$$

where $dU_t > 0$ only when $x_t \leq -X^*$, $dD_t > 0$ only when $x_t \geq X^*$.

Suppose the cost per unit of the intervention is κ , equal for both buying and selling reserves. The monetary policy authority minimizes the discounted sum of the squared deviation of the exchange rate from a target value, which is usually normalized to 0 (so we impose symmetry), and proportional intervention costs. Then the value function of the monetary policy authority for the corresponding problem will be

$$v(x) = \min_{U,D} \mathbb{E} \left[\int_0^\infty e^{-\rho t} s_t^2 dt + \kappa \int_0^\infty e^{-\rho t} (dU_t + dD_t) \middle| x_0 = x \right] \quad (13)$$

The value function satisfies a quasi-variational inequality system. In the inaction region $x \in (-X^*, X^*)$

$$\rho v(x) = f^2(x) + v''(x) \frac{\sigma^2}{2},$$

where $f(x) = s$ is the exchange rate given by (5). At each boundary, the value of remaining cannot exceed the value of intervening:

$$v(x) \leq \min_{\xi} \{v(\xi) + \kappa |\xi - x|\}.$$

The optimality of the intervention trigger X^* is characterized by four conditions at the boundaries. The smooth-pasting conditions,

$$v'(-X^*) = -\kappa \quad v'(X^*) = \kappa,$$

ensure that the marginal value of intervention equals its marginal cost, ruling out discrete jumps in the intervention rate. Two conditions for local linearity of the value function (super-contact):

$$v''(-X^*) = 0 \quad v''(X^*) = 0,$$

pin down the optimal location of the trigger itself.

Steady-state reformulation The steady-state formulation abstracts from initial conditions and focuses on the long-run average behavior of the system. The steady-state analogue of (13) is

$$\min_X \text{Var}(s) + \kappa(A_u(X) + A_d(X)), \quad (14)$$

where $A_u(X) := \lim_{t \rightarrow \infty} \mathbb{E}[U_t]/t$ and $A_d(X) := \lim_{t \rightarrow \infty} \mathbb{E}[D_t]/t$ denote the expected long-run rates at which U_t and D_t accumulate – that is, the expected rate of accumulation of local time at each boundary.

Lemma 2 (Local-Time Accumulation at the Boundaries). *Under the dynamics $dx_t = \sigma dW_t + dU_t - dD_t$ with reflecting barriers at $-X$ and X , and stationary density $m(x; X) = \frac{1}{2X}$,*

$$A_u(X) = A_d(X) = \frac{\sigma^2}{4X}.$$

By Lemma 2, the total intervention cost rate is

$$\kappa(A_u(X) + A_d(X)) = \frac{\kappa\sigma^2}{2X}.$$

This expression is intuitive: higher fundamental volatility σ^2 increases the rate at which local time accumulates at each boundary, while a wider band reduces it. Substituting the exact variance (10) and the intervention cost, problem (14) becomes

$$\min_X \frac{1}{2X\beta^2} \left(\frac{2X^3}{3} - \frac{4X}{\eta^2} + \frac{5 \tanh(\eta X)}{\eta^3} - \frac{X}{\eta^2 \cosh^2(\eta X)} \right) + \frac{\kappa\sigma^2}{2X}. \quad (15)$$

Both terms are well-behaved: the variance term is strictly increasing in X , while the intervention cost is strictly decreasing and convex. The two terms share units of squared exchange rate deviations, making the trade-off directly interpretable. The optimal band balances these forces at the margin.

Closed-Form Solutions All optimization problems considered below take the form

$$\min_X \text{Var}(X) + \frac{\kappa\sigma^2}{2X},$$

where $\text{Var}(X)$ is increasing and $\kappa\sigma^2/(2X)$ is strictly convex and decreasing. The derivative of the objective function is strictly increasing, implying that the problem has at most one critical point.

For the two closed-form approximations, strict convexity of the full objective – and hence uniqueness of the interior minimizer – is established directly below by computing the second derivative and verifying it is strictly positive for all $X > 0$. For the exact objective, the analogous second-order condition reduces to the sign of a single function $\psi(t)$, defined in the proof below; we do not pursue a fully analytic proof that $\psi(t) > 0$ for all $t > 0$, and instead verify it numerically¹. The following proposition summarizes exact and approximated solutions.

Proposition 3 (Optimal Currency Bands: Symmetric Case). *Under a symmetric driftless fundamental and uniform stationary distribution, the optimal band problem (15) has a unique solution in each case:*

1. **Exact solution.** *The optimal fundamental boundary X solves*

$$\frac{2X}{3\beta^2} - \frac{5 \tanh(\eta X)}{2\eta^3 X^2 \beta^2} + \frac{5}{2\eta^2 X \beta^2 \cosh^2(\eta X)} + \frac{\sinh(2\eta X)}{2\eta \beta^2 \cosh^4(\eta X)} - \frac{\kappa \sigma^2}{2X^2} = 0. \quad (16)$$

2. **Narrow-band approximation** (X is small):

$$X = \left(\frac{105}{272}\right)^{1/7} \kappa^{1/7} \sigma^{6/7} \quad b = \frac{2}{3} \left(\frac{105}{272}\right)^{3/7} \kappa^{3/7} \sigma^{4/7}. \quad (17)$$

3. **Wide-band approximation** (X is large):

$$X = \left(\frac{3\beta^2}{4}\right)^{1/3} \kappa^{1/3} \sigma^{2/3} \quad b = \left(\frac{3}{4\beta}\right)^{1/3} \kappa^{1/3} \sigma^{2/3}. \quad (18)$$

Before formalizing the result, it is useful to note a scaling property of the model. Both sources of costs, the variance of the exchange rate and intervention costs, depend on the relative magnitude of the band X compared to fundamental volatility σ . This suggests that what matters for optimal policy is not the absolute level of X , but its size relative to the scale of shocks. As a result, the optimal band is expected to exhibit a homogeneity property, depending only on the ratio of intervention costs to volatility.

Corollary 2 (Homogeneity). *The optimal band X^* is homogeneous of degree 1 in (κ, σ) ; or equivalently, X^*/σ depends only on the ratio κ/σ , that is*

$$v := X^*/\sigma = \phi(\kappa/\sigma). \quad (19)$$

1. *In fact, for the exact solution $\phi(\kappa/\sigma) = g^{-1}(\kappa/\sigma)$, with*

$$g(v) = \frac{4v^3}{3\beta^2} - \frac{5 \tanh(\sqrt{2\beta}v)}{2\sqrt{2}\beta^{7/2}} + \frac{5v}{2\beta^3 \cosh^2(\sqrt{2\beta}v)} + \frac{\sqrt{2}v^2 \sinh(2\sqrt{2\beta}v)}{2\beta^{5/2} \cosh^4(\sqrt{2\beta}v)}.$$

2. *For the narrow-band case (X is small):*

$$\phi(\kappa/\sigma) = \left(\frac{105}{272} \frac{\kappa}{\sigma}\right)^{1/7}$$

¹ $\psi(t) \rightarrow 0$ as $t \rightarrow 0^+$ and $\psi(t) \rightarrow 2$ as $t \rightarrow \infty$, remaining strictly positive throughout; see Figure 10 in Appendix A.3.

3. For the wide-band case (X is large):

$$\phi(\kappa/\sigma) = \left(\frac{3\beta^2 \kappa}{4 \sigma} \right)^{1/3}$$

Comparative statics Both approximations predict wider bands when intervention costs κ or fundamental volatility σ^2 are higher. The economic logic is straightforward: higher volatility causes the fundamental to reach the boundary more often, so the authority finds it cheaper to widen the band; higher intervention costs directly raise the price of each boundary encounter, pushing in the same direction. The relevant comparison across regimes uses elasticities, which are unit-free. Let $\varepsilon_{\sigma^2}(b) := \frac{\partial b}{\partial \sigma^2} \frac{\sigma^2}{b}$ and $\varepsilon_{\kappa}(b) := \frac{\partial b}{\partial \kappa} \frac{\kappa}{b}$. These are reported in Table 1.

Table 1: Elasticity of the optimal band b with respect to fundamental variance σ^2 and intervention cost κ .

	$\varepsilon_{\sigma^2}(b)$	$\varepsilon_{\kappa}(b)$
Narrow bands	2/7	3/7
Wide bands	1/3	1/3

Two features of Table 1 stand out. First, in the narrow-band regime intervention costs dominate: a 1% increase in κ widens the band by 3/7%, against only 2/7% for an equal increase in σ^2 . When the band is already tight, intervention is frequent and even small cost increases have an outsized effect on the optimal width. Second, in the wide-band regime both elasticities converge to 1/3, reflecting the diminishing importance of the cost channel as intervention becomes rare. The crossover – cost elasticity falling from 3/7 to 1/3 as the band widens, variance elasticity rising from 2/7 to 1/3 – captures the changing structure of the policy trade-off: tight bands are primarily sustained by low intervention costs, wide bands by high fundamental volatility.

Since the optimal band b^* is implicitly defined by (16), the exact elasticities do not admit closed-form expressions. However, because the narrow- and wide-band approximations are limiting cases of the same problem, the exact elasticities lie between their limiting values for all intermediate band widths.

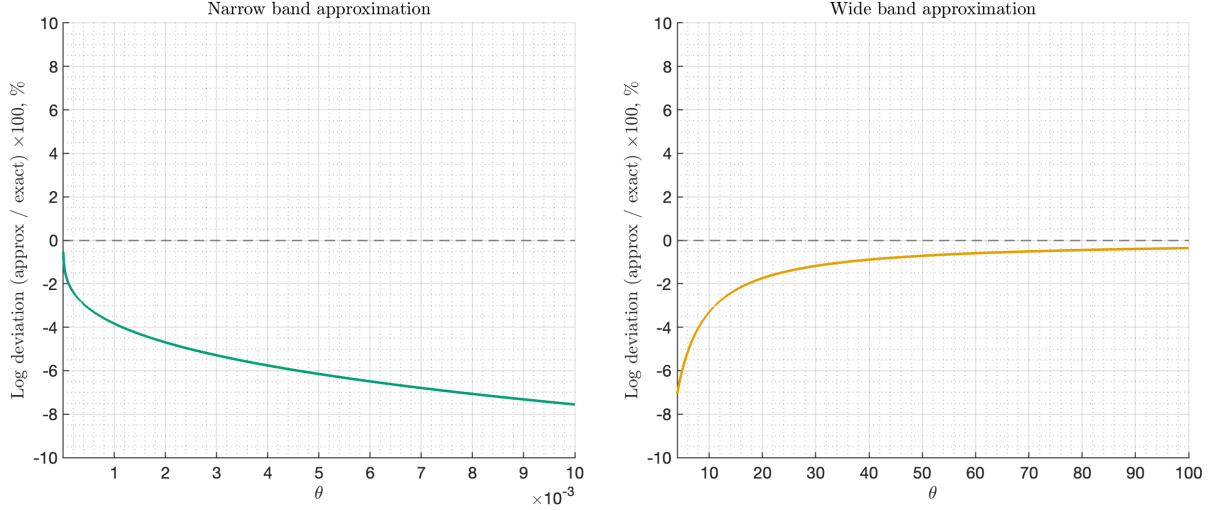


Figure 2: Relative error of the two band approximations as a function of $\theta := \kappa/\sigma$, measured as $100 \times \log(X_{approx}/X_{exact})$. Left: narrow-band approximation; right: wide-band approximation.

Figure 2 plots the log deviation of each approximation from the exact solution as a function of $\theta := \kappa/\sigma$. Several features are noteworthy. First, both approximations always understate X^* . Second, small θ , and subsequently small v , translates into narrow band, because g is increasing with $g \sim v^7$ near 0. Third, the two approximations coincide exactly at $\theta \approx 0.498$, where both understate X^* by about 25%; the calibrated values reported in Section 3.3 ($\theta \approx 0.47$ in both ERM I regimes) lie almost exactly at this crossover, explaining why neither closed-form approximation is reliable for the empirical application and motivating the numerical solution of the exact first-order condition (16).

2.3 Modeling Assumptions and Their Empirical Relevance

The model developed above rests on four assumptions that are convenient for obtaining closed-form and exact results, but that the Italian ERM I experience violates to varying degrees. We list them here, in advance of the empirical application, so that the direction in which each violation is likely to bias the calibration is stated once and referenced rather than discovered piecemeal in the results.

1. Full credibility. The exact variance (10) and optimality condition (16) assume zero probability of realignment at any horizon. The lira violates this directly – Table 2 records more than ten central-parity adjustments between 1979 and 1992 – and anticipated realignments introduce drift toward the depreciation boundary ahead of each one, breaking the honeymoon effect’s no-arbitrage condition (Bertola and Caballero, 1992; Svensson, 1991). Choosing realignment-free estimation windows addresses this only to first order, since anticipation of the realignment at each window’s end can still generate drift within it. Because $\hat{\sigma}$ is identified from the observed variance of s_t , any such excess variance is absorbed into $\hat{\sigma}$ and, via the

first-order condition, into $\hat{\kappa}$; since the optimal band is increasing in both, this mechanically inflates the model’s implied band relative to one with an explicit realignment-risk term.

2. Zero drift. This is a special case of credibility: even absent a discrete jump, anticipation of future policy action can tilt the fundamental’s conditional mean toward one boundary. We choose the two estimation windows (12 January 1987–4 January 1990 and 5 January 1990–11 September 1992) specifically to avoid actual realignments within each window, which addresses the stationarity concern to first order; it does not, however, remove anticipation of a realignment at the end of the window, which can still generate drift inside it.
3. Symmetric intervention cost. The model assumes a single κ for both boundaries (13), but defending the depreciation boundary draws down a finite reserve stock – growing costlier as it depletes – while defending the appreciation boundary faces no such constraint. The lira spent nearly the entire narrow-band period pressed against the depreciation boundary (Section 3.3), so $\hat{\kappa}$ is identified almost entirely from the costlier side; fitting one symmetric cost to this one-sided pattern mechanically overstates the true average. This is a sharper reading of the gap with Miller and Zhang (1996)’s benchmark than “persistent pressure” alone: a model that pools two different costs into one overstates the lower whenever the data are dominated by the higher.
4. Free capital mobility. The model’s asset-pricing relation assumes frictionless UIP. Italy’s capital controls, progressively liberalized through the late 1980s, may have weakened UIP-based intervention incentives in the earlier $\pm 6\%$ sub-window; we do not quantify this separately.

Items 1–3 inflate $\hat{\kappa}$ (and, for item 1, $\hat{\sigma}$ as well) relative to a correctly specified model, which in turn inflates the model’s implied optimal bands. The results below should accordingly be read as an upper bound on what a model with full credibility, zero drift, and boundary-specific costs would imply for the lira’s band width.

3 Quantitative Application: Italy in the ERM I

3.1 Institutional Setting and Data

The Exchange Rate Mechanism This section provides a quantitative evaluation of optimal currency bands using the experience of the Exchange Rate Mechanism (ERM I) prior to the adoption of the euro. The empirical application focuses on Italy, whose participation in the ERM I offers a particularly informative case due to the presence of both wide and narrow bands. Our main estimation sample covers the period 1987–1992, often described as a phase of relatively strong policy commitment (e.g. [Detragiache and Hamann, 1999](#)), with reduced inflation and narrower fluctuations.

Under ERM I, each currency maintained a central parity against the European Currency Unit (ECU) within a symmetric fluctuation band, with central banks required to intervene at the edges. In the language of target zone models, these bands correspond to reflecting barriers, and occasional consensus realignments of central parities correspond to discrete shifts in the center of the zone.

Italian lira dynamics In particular, Italy entered the EMS in March 1979 with a $\pm 6\%$ band around the lira’s central parity (other members had $\pm 2.25\%$). Throughout the 1980s, the lira was managed with the objective of disinflation, with policy effectively anchored to the ECU. Over time, as inflation declined, Italy transitioned to the narrow $\pm 2.25\%$ band in January 1990, marking a shift toward stronger exchange rate commitment. Figures 3 and 4 illustrate the full ERM I history of the lira and zoom into the 1987–1992 estimation window, respectively. The corresponding series in levels (Italian lira per ECU) are reproduced in Appendix A.3 for reference.

However, this commitment must be interpreted with caution. The Italian experience was characterized by frequent parity realignments: between 1979 and 1992, the lira underwent more than ten adjustments of its central parity (more precisely, system-wide realignments). These realignments were typically devaluations aimed at restoring competitiveness. Table 2 summarizes these episodes and illustrates that the ERM I was not a fixed target zone, but rather a sequence of regimes with shifting centers.

A critical episode occurred in September 1992. On 14 September, following intense speculative pressure, the Italian authorities announced a devaluation of the lira within the ERM I. This move triggered further market instability rather than restoring confidence. On 16 September 1992, coinciding with the sterling crisis and commonly referred to as “Black Wednesday”, Italy suspended ERM I participation and allowed the lira to float freely. Italy remained outside the ERM I until 24 November 1996, when it rejoined under the common $\pm 15\%$ band that had been adopted for all members following the 1993 EMS crisis.

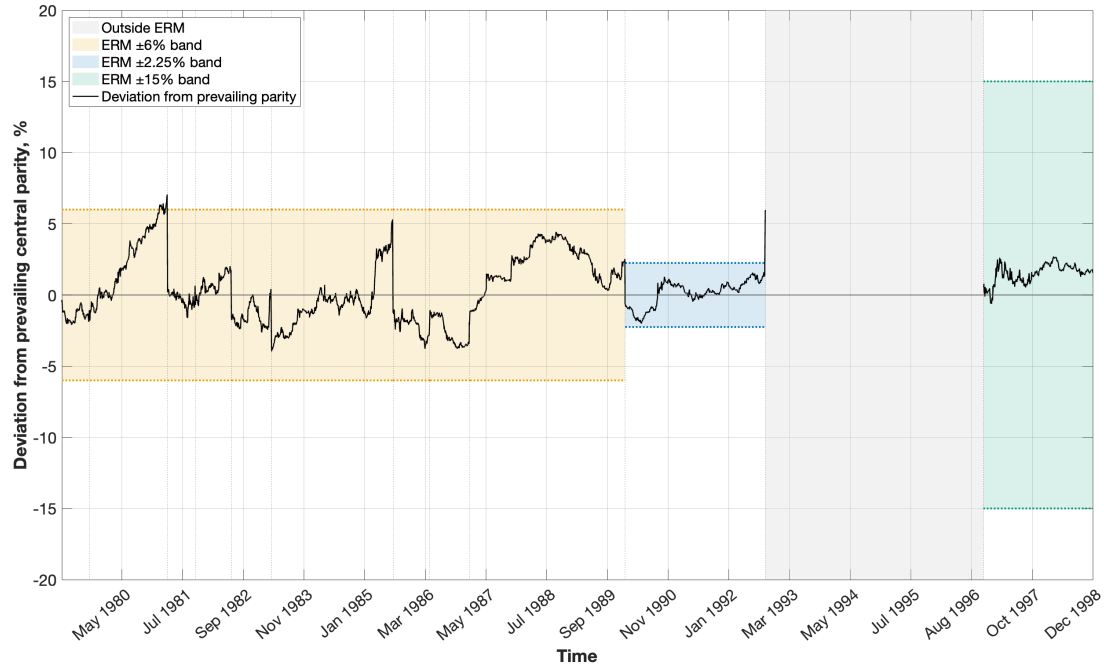


Figure 3: Italian lira's deviation from its prevailing central parity (%) over the 1987–1992 estimation window (12 January 1987 – 16 September 1992), spanning the transition from the $\pm 6\%$ to the $\pm 2.25\%$ ERM I band in January 1990 (marked by the dotted vertical line). Shaded orange and blue bands show the $\pm 6\%$ and $\pm 2.25\%$ fluctuation margins respectively; the solid black line is the lira's deviation from the prevailing central parity.

Source: Based on ECB daily data.

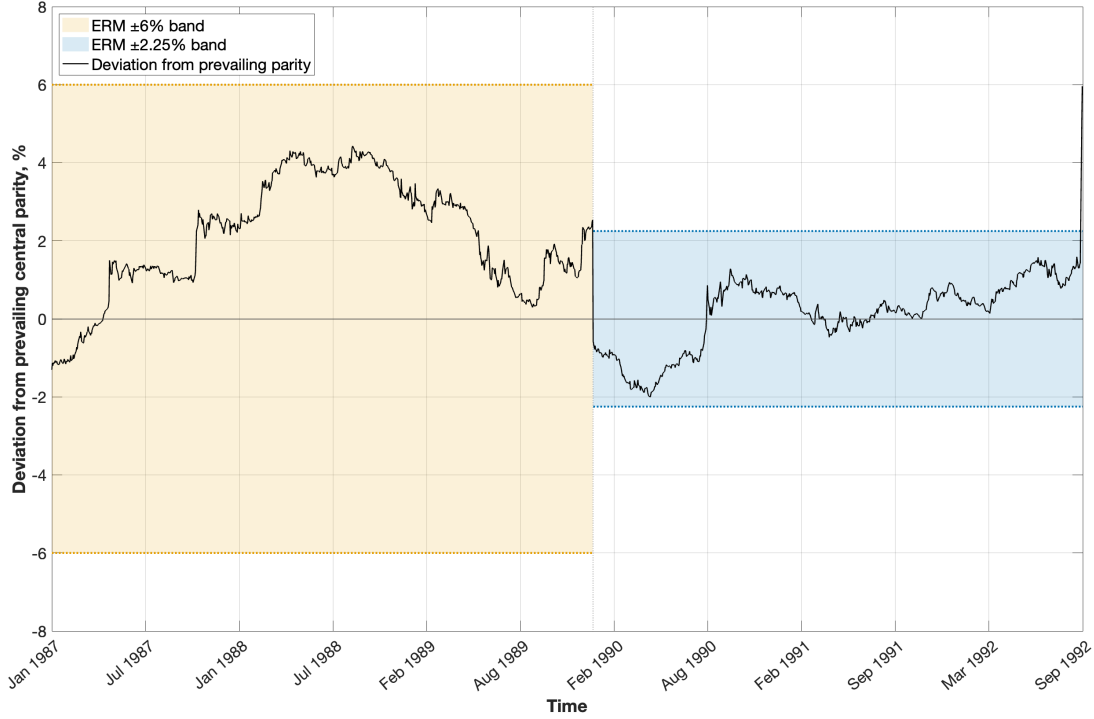


Figure 4: Italian lira's deviation from its prevailing central parity (%), March 1979 – December 1998, across the $\pm 6\%$, $\pm 2.25\%$, outside-ERM I, and $\pm 15\%$ regimes. Shaded bands show each regime's fluctuation margin, stepping to a new band width and/or central parity at every realignment (marked by dotted vertical lines); the solid black line is the lira's actual deviation from the parity prevailing at each date, and the horizontal line at zero marks the central parity itself.

Source: Based on ECB daily data.

Table 2: Italian lira central rates in EMS/ERM I and devaluation steps.

Date	Action	Central parity (lira per ECU)	Δ vs. prior, %
12.03.1979	Entry into EMS (broad band $\pm 6\%$)	1148.15	–
24.09.1979	EMS realignment (no lira change)	1159.42	+0.98%
22.03.1981	Realignment (lira -6.0%)	1262.92	+8.93%
05.10.1981	Realignment (lira -3.0%)	1300.67	+2.99%
14.06.1982	Realignment (lira -2.75%)	1350.27	+3.81%
21.03.1983	Realignment (lira -2.5%)	1386.78	+2.70%
20.07.1985	Realignment (lira -6.0%)	1520.06	+9.61%
07.04.1986	EMS realignment (no lira change)	1496.21	-1.57%
12.01.1987	EMS realignment (German mark +3%)	1483.58	-0.84%
05.01.1990	Peg moved to $\pm 2.25\%$ (lira -3.7%)	1529.70	+3.11%
24.11.1996	Re-entry into ERM at $\pm 15\%$ bands	1906.48	+24.63%
01.01.1999	Euro adoption (ERM-II launch)	–	–

Source: Official EU/Commission records (EC Bulletins and Council Press Releases) provide each adjustment.

Several features of Table 2 are directly relevant to the empirical strategy. First, the frequency of realignments implies that agents operating within the ERM I likely attached positive probability to discrete parity changes at any point in time. As Bertola and Caballero (1992) and Svensson

(1991) show, such anticipations break the standard no-arbitrage condition underlying simple target zone models and introduce drift toward the depreciation boundary in the pre-realignment period. This is the primary credibility caveat for the empirical application.

Second, the system-wide realignment of January 1987, in which the Deutsche Mark was revalued without a unilateral Italian policy action, provides a natural starting point for the estimation window. Because no lira-specific parity adjustment occurred after this date until January 1990, the period 12 January 1987 – 4 January 1990 constitutes a clean within-regime window free of central-parity shifts, making it the most suitable sub-sample for identifying within-band dynamics under the $\pm 6\%$ regime.

Third, the 1992 crisis establishes the end of the narrow-band window. The three days of speculative attack (14–16 September 1992) are excluded from estimation, as the extreme volatility and discontinuous reserve movements of that episode would distort the identification of within-band dynamics. The second sub-sample therefore closes on 11 September 1992.

Data We use European Central Bank (ECB) daily data on the Italian lira-ECU bilateral rate and Italian short-term interest rates, covering 7 March 1979 to 31 December 1998. Observations are available on weekdays only. For the calibration we distinguish two sub-samples: the $\pm 6\%$ band period (12 January 1987 – 4 January 1990) and the $\pm 2.25\%$ band period (5 January 1990 – 11 September 1992).

For the semi-elasticity of money demand α we follow Coenen and Vega (2001), whose estimates are based on quarterly euro-area data covering 1980:Q4–1998:Q4. Their baseline estimate of $\alpha = 0.87$, which converts into $\beta := 1/\alpha = 1.1494$, is used throughout. Exchange-rate variance $\text{Var}(s)$ is computed as the sample variance within each sub-window.

3.2 Calibration Strategy

Throughout this section, every structural estimate – $\hat{\sigma}$, $\hat{\kappa}$, and the implied optimal band – is recovered by solving the model’s exact equations (6), (10), and (16) numerically, not the narrow- or wide-band closed-form approximations of Section 2. Those approximations remain the theoretical contribution, delivering transparent comparative statics in the two limiting regimes; the calibration below is a separate, numerically exact exercise that does not inherit their approximation error. One place later in this section does invoke a closed-form relationship for intuition rather than for computation – the narrow-band scaling $b \propto \kappa^{3/7} \sigma^{4/7}$ used to interpret the α -robustness check – and is flagged explicitly below as a heuristic reading of exact-equation results, not as the mechanism generating them.

Algorithm We observe exchange-rate data s_t within a band of known width b_{obs} . Treating β as a known structural parameter, the model has three unknowns: σ , the volatility of the fundamental; X , the half-width of the intervention trigger; and κ , the intervention cost parameter. The identification strategy exploits the fact that Italy operated under two distinct band regimes, so that

we can apply the same procedure to each regime independently and recover a pair of structural estimates $(\hat{\sigma}, \hat{\kappa})$ for each.

The procedure consists of two steps. First, we solve the inverse problem: given the observed band width b_{obs} and the sample variance $\text{Var}(s)$ of the exchange rate, we back out the structural parameters $(\hat{\sigma}, \hat{X}, \hat{\kappa})$ that are consistent with these moments. Second, we conduct a counterfactual analysis: using the κ estimate recovered from one regime, we ask what band the planner would have chosen optimally had the fundamental volatility been that of the other regime.

The counterfactual exercise has a natural interpretation as a decomposition. Because the optimal band b^* depends on both σ and κ , two distinct channels could in principle account for the observed change in band width across regimes: (i) a change in σ , reflecting a shift in macroeconomic conditions, and (ii) a change in κ , reflecting a change in the structural cost of intervention. By combining $\hat{\kappa}$ from one regime with $\hat{\sigma}$ from the other, we isolate each channel's contribution. If the resulting counterfactual bands are close to the observed widths, the structural parameters alone can rationalize the regime change. If not, some combination of non-structural motives must be invoked.

Three equations pin down the three unknowns (σ, X, κ) :

1. Boundary $b(X, \sigma)$ from (6):

$$b = \frac{1}{\eta\beta} (\eta X - \tanh(\eta X))$$

2. Variance $\text{Var}(X, \sigma)$ from (10):

$$\text{Var}(s) = \frac{1}{2X\beta^2} \left(\frac{2X^3}{3} - \frac{4X}{\eta^2} + \frac{5 \tanh(\eta X)}{\eta^3} - \frac{X}{\eta^2 \cosh^2(\eta X)} \right)$$

3. Optimality condition from (16):

$$\frac{2X}{3\beta^2} - \frac{5 \tanh(\eta X)}{2\eta^3 X^2 \beta^2} + \frac{5}{2\eta^2 X \beta^2 \cosh^2(\eta X)} + \frac{\sinh(2\eta X)}{2\eta \beta^2 \cosh^4(\eta X)} = \frac{\kappa \sigma^2}{2X^2}$$

While the first two hold for any X , the last one is implied by the optimal choice of X in problem (14). Intuitively: the boundary equation pins down X , the variance equation pins down σ , and the optimality condition then identifies κ .

To make the system computationally tractable, we exploit a homogeneity property (19) of the equations. Defining $v := X/\sigma$ and $\theta := \kappa/\sigma$, the equations (6), (10), and (16) reduce to:

1. Boundary $b(v, \sigma)$:

$$b = \sigma h(v) \quad \text{with} \quad h(v) = \frac{v}{\beta} - \frac{\tanh(\sqrt{2\beta}v)}{\beta\sqrt{2\beta}} \quad (20)$$

2. Variance $\text{Var}(v, \sigma)$:

$$\text{Var}(s) = \sigma^2 q(v) \quad \text{with} \quad q(v) = \frac{v^2}{3\beta^2} - \frac{1}{\beta^3} + \frac{5 \tanh(\sqrt{2\beta}v)}{4\beta^3 \sqrt{2\beta}v} - \frac{1}{4\beta^3 \cosh^2(\sqrt{2\beta}v)} \quad (21)$$

3. Optimality condition depending on (X, σ, κ) :

$$g(v) = \theta \quad \text{with} \quad g(v) = \frac{4v^3}{3\beta^2} - \frac{5 \tanh(\sqrt{2\beta}v)}{2\sqrt{2}\beta^{7/2}} + \frac{5v}{2\beta^3 \cosh^2(\sqrt{2\beta}v)} + \frac{\sqrt{2}v^2 \sinh(2\sqrt{2\beta}v)}{2\beta^{5/2} \cosh^4(\sqrt{2\beta}v)} \quad (22)$$

Dividing the variance equation (21) by the square of the boundary equation (20) eliminates σ , yielding a single equation in v alone:

$$\frac{\text{Var}(s)}{b_{\text{obs}}^2} = \frac{q(v)}{h(v)^2} \iff b_{\text{obs}}^2 q(v) - \text{Var}(s) h(v)^2 = 0.$$

Since $q(v)/h(v)^2$ is monotone for $v > 0$ ², this equation has a unique solution $\hat{v}$, which is found numerically. Given $\hat{v}$, we recover $\hat{\sigma} = b_{\text{obs}}/h(\hat{v})$, and then $\hat{\kappa} = \hat{\sigma} g(\hat{v})$, which follows directly from equation (22).

The inverse problem is solved independently on each regime, yielding estimates $(\hat{\sigma}_{2.25}, \hat{\kappa}_{2.25})$ and $(\hat{\sigma}_6, \hat{\kappa}_6)$. To conduct the counterfactual, we take $\hat{\kappa}$ from one regime and ask: given this cost of intervention and the fundamental volatility $\hat{\sigma}$ observed in the other regime, what band would the planner have chosen? Formally, the counterfactual optimal band b^* solves the planner's optimality condition for a given pair $(\hat{\sigma}, \hat{\kappa})$:

$$g(v^*) = \frac{\hat{\kappa}}{\hat{\sigma}}, \quad b^* = \hat{\sigma} \cdot h(v^*).$$

This is a direct computation: given the cross-regime $(\hat{\sigma}, \hat{\kappa})$ pair, we solve for v^* numerically and recover b^* immediately. No iteration is required. We perform this calculation twice (once using $(\hat{\sigma}_6, \hat{\kappa}_{2.25})$ and once using $(\hat{\sigma}_{2.25}, \hat{\kappa}_6)$) to quantify the separate contributions of volatility and cost to the observed difference in band widths across regimes.

3.3 Results

Table 3 reports the baseline calibration results. The estimated fundamental volatilities $\hat{\sigma}_6 = 0.0612$ and $\hat{\sigma}_{2.25} = 0.0362$ are smaller than the parameter values used in the numerical exercises of Miller and Zhang (1996) and Im (2001), who explore σ^2 in the range 0.01–0.3, consistent with our data being at daily frequency.

The estimated intervention costs diverge sharply from those benchmarks. Miller and Zhang (1996) set their unit intervention cost c in the range 2×10^{-3} – 8×10^{-3} , whereas our estimates $\hat{\kappa}_6 = 0.0289$ and $\hat{\kappa}_{2.25} = 0.0171$ are roughly two to fifteen times larger, depending on the regime and where in Miller and Zhang's range the comparison is drawn. We attribute this to the chronic one-sided pressure the lira faced against the depreciation boundary throughout the narrow-band period (Figure 4): intervention was a near-continuous commitment rather than the occasional discrete event assumed in stylized numerical exercises, and the large $\hat{\kappa}$ absorbs this burden.

²Figures 8 and 9 in the Appendix A.3 plot the auxiliary functions $q(v)$, $h(v)$, and their derivatives, confirming the monotonicity properties that guarantee a unique solution.

The calibrated values of $\theta = \kappa/\sigma$ are approximately 0.47 in both regimes, corresponding to $\eta X \approx 1.52$ – an intermediate value, far from both the $\eta X \rightarrow 0$ narrow-band limit and the $\eta X \rightarrow \infty$ wide-band limit underlying the two closed-form approximations in Figure 2. Figure 12 in Appendix A.3 shows both approximation curves together on a single $\log\theta$ axis, with the calibrated $\theta \approx 0.47$ marked directly, making this intermediate position visually explicit. At this calibrated value, neither approximation is reliable: the narrow-band approximation understates the exact optimal band by about 24%, and the wide-band approximation by about 25%, with the narrow-band formula only marginally the less inaccurate of the two. This does not affect the identification strategy itself, however, since the calibration solves the exact equations directly rather than either approximation (see the opening of this section).

Table 3: Calibration results.

	$\pm 6\%$ regime	$\pm 2.25\%$ regime
	12.01.1987–04.01.1990	05.01.1990–11.09.1992
	$N = 760$	$N = 681$
Observed band b_{obs}	6.00%	2.25%
Observed variance $\widehat{Var}(s)$	2.17% ²	0.76% ²
Estimated X^*	6.12%	3.62%
Estimated $\hat{\sigma}$	0.0612	0.0362
Estimated $\hat{\kappa}$	0.0289	0.0171
$v = X^*/\hat{\sigma}$	1.0003	1.0002
$\theta = \hat{\kappa}/\hat{\sigma}$	0.4720	0.4717

Notes: N is the number of daily (weekday) observations within each sub-sample, excluding the three speculative-attack days (14–16 September 1992) from the $\pm 2.25\%$ window. All exchange-rate series are at daily frequency, and $\hat{\sigma}$ is a daily volatility of the fundamental in the same percentage units as b_{obs} . $\hat{\kappa}$ is the model’s reduced-form intervention-cost parameter, calibrated jointly with $\hat{\sigma}$ via the first-order condition (eq. 16) and reported in the model’s own internal units; following standard practice in the target-zone literature (e.g. Krugman 1991), it is not assigned an independent economic unit.

The counterfactual exercise decomposes the change in the optimal trigger across regimes into two structural channels. The baseline optimal trigger falls from $X^* = 6.12\%$ under the $\pm 6\%$ regime to $X^* = 3.62\%$ under the $\pm 2.25\%$ regime, a decline of 2.50 percentage points. Holding the intervention cost at its wide-band value $\hat{\kappa}_{6\%} = 0.0289$ and replacing fundamental volatility with its narrow-band value $\sigma_{2.25\%} = 0.0362$ yields $X^* = 4.07\%$; the volatility channel therefore accounts for 2.05pp of the decline. Holding volatility at $\hat{\sigma}_{6\%} = 0.0612$ and replacing the cost with $\hat{\kappa}_{2.25\%} = 0.0171$ yields $X^* = 5.47\%$; the cost channel accounts for 0.65pp. Both channels are active, and the volatility channel is the larger of the two.

The two partial effects sum to 2.70pp against a total change of 2.50pp, so they overlap rather than add, and neither on its own reproduces the narrow-band trigger. The volatility channel’s dominance follows directly from the counterfactual computation itself rather than from the narrow- or wide-band asymptotic elasticities in Table 1: at the calibrated $\theta \approx 0.47$ the model sits between

both asymptotic regimes, so neither the narrow-band elasticities (2/7, 3/7) nor the wide-band ones (1/3, 1/3) apply exactly, and the exact local elasticities at this point are not available in closed form. Neither counterfactual, nor the two combined, brings the trigger down to the $\pm 2.25\%$ width actually announced in 1990, pointing to motives beyond cost minimisation: commitment value, coordination of market expectations, and a precautionary buffer against speculative attacks (Krugman, 1979; Bertola and Caballero, 1992).

Table 4: Baseline Calibration and Counterfactual Optimal Bands.

	Baseline		Counterfactual	
	$b = 6\%$	$b = 2.25\%$	$\hat{\kappa}_{6\%}, \hat{\sigma}_{2.25\%}$	$\hat{\kappa}_{2.25\%}, \hat{\sigma}_{6\%}$
Observed band b_{obs}	6.00%	2.25%	—	—
Observed variance $\widehat{\text{Var}}(s)$	2.17% ²	0.76% ²	—	—
$\hat{\sigma}$	0.0612	0.0362	0.0362	0.0612
$\hat{\kappa}$	0.0289	0.0171	0.0289	0.0171
$\theta = \hat{\kappa}/\hat{\sigma}$	0.4720	0.4717	0.7990	0.2786
$v = X^*/\hat{\sigma}$	1.0003	1.0002	1.1260	0.8944
X^*	6.12%	3.62%	4.07%	5.47%

Notes: Counterfactual columns fix one structural parameter from each baseline regime and substitute the other; unlike the baseline columns, they are not estimated from any sample and so have no associated N or observed moments. The baseline columns reproduce the two samples of Table 3 ($N = 760$ and $N = 681$ daily observations respectively); $\hat{\kappa}$ and $\hat{\sigma}$ are reported in the same units as in Table 3.

Figure 5 plots the optimal intervention trigger X^* as a function of fundamental volatility σ under each of the two regimes κ estimates, holding $\beta = 1.1494$ fixed throughout. First, X^* is increasing in σ under both parameterizations: higher fundamental volatility calls for a wider intervention band, as the planner tolerates larger swings before intervening to avoid excessive costs. Second, the curve corresponding to $\hat{\kappa}_{2.25\%} = 0.0171$ lies below that for $\hat{\kappa}_{6\%} = 0.0289$ at every σ : a higher intervention cost leads the planner to intervene later, so the planner sets a wider trigger. Finally, the two vertical dashed lines mark the estimated fundamental volatilities $\hat{\sigma}_{2.25\%} = 0.0362$ and $\hat{\sigma}_{6\%} = 0.0612$. Reading off the figure, the green curve at $\hat{\sigma}_{2.25\%} = 0.0362$ and the yellow curve at $\hat{\sigma}_{6\%} = 0.0612$ give the respective cross-regime optimal triggers, which translate into the optimal band widths reported in Table 4.

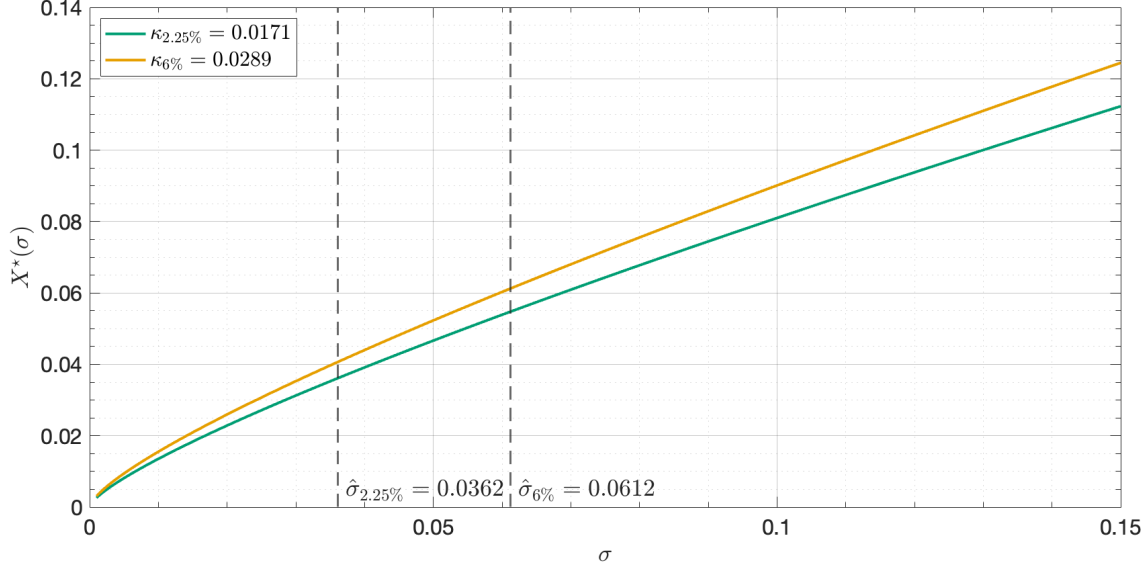


Figure 5: Optimal intervention trigger X^* as a function of fundamental volatility σ , for $\hat{\kappa}_6 = 0.0289$ (green) and $\hat{\kappa}_{2.25} = 0.0171$ (yellow), with $\beta = 1.1494$. Vertical dashed lines mark the estimated fundamental volatilities $\hat{\sigma}_{2.25\%} = 0.0362$ and $\hat{\sigma}_{6\%} = 0.0612$.

The semi-elasticity α (equivalently $\beta = 1/\alpha$) is not estimated, so we re-solve the inverse problem over a range of α to estimate its influence on the recovered parameters (Figure 13). As α rises, the inverse problem attributes a larger share of the observed band to intervention cost and a smaller share to fundamental volatility: $\hat{\kappa}$ increases and $\hat{\sigma}$ decreases in both regimes. Because the two move in opposite directions, the recovered ratio κ/σ is not stable: it rises with α . The implied optimal band, however, is far less sensitive. A useful heuristic for why – though Figure 14 itself is generated from the exact equations, not from this approximation – comes from the narrow-band closed form $b \propto \kappa^{3/7} \sigma^{4/7}$: the decline in $\hat{\sigma}$ should partly offset the rise in $\hat{\kappa}$, so the optimal trigger should move much less than $\hat{\kappa}$ alone, which is indeed the pattern Figure 14 shows. What is sensitive to α is the level of $\hat{\kappa}$, which is why the comparison with the numerical exercises of Miller and Zhang (1996) and Im (2001) is best read at a common semi-elasticity.

4 Conclusion

This work develops an analytically tractable framework for optimal exchange rate band design, calibrates it to the Italian ERM I experience, and draws lessons for the structural determinants of observed band widths.

Theoretical contributions On the theoretical side, three results stand out. First, building on [Krugman's \(1991\)](#) target zone model, we derive an exact closed-form expression for the steady-state variance of the exchange rate, which admits two complementary approximations: $\text{Var}(s) \approx (17/35)b^2$ for narrow bands and $\text{Var}(s) \approx (1/3)b^2$ for wide bands. Both regimes deliver a clean proportionality $\text{Var}(s) \propto b^2$, with the constant $17/35 > 1/3$ implying that exchange rate variance is higher under a narrow-band regime than a wide-band one for any given width b .

Second, the optimal band problem (minimizing exchange rate variance plus proportional intervention costs) has a unique interior solution in each approximation regime. The narrow-band solution scales as $b^* \propto \kappa^{3/7}\sigma^{4/7}$ and the wide-band solution as $b^* \propto \kappa^{1/3}\sigma^{2/3}$; the exact first-order condition (16) pins down the solution numerically. The elasticity analysis (Table 1) reveals a systematic asymmetry: in the narrow-band regime, intervention costs are the dominant driver (elasticity 3/7), while volatility plays a smaller role (elasticity 2/7); in the wide-band regime both elasticities converge to the common value 1/3. As the optimal band widens, cost elasticity falls and variance elasticity rises – a crossover that reflects the changing structure of the policy trade-off.

Third, the homogeneity result (Corollary 2) shows that the optimal band depends on the ratio κ/σ alone, not on the levels separately. This normalization reduces the dimensionality of the parameter space and makes cross-country and cross-regime comparisons interpretable in one dimension: economies with high intervention costs relative to fundamental volatility optimally choose wider bands.

Quantitative findings The cross-regime calibration using Italian ERM I data yields two quantitative findings, reported in Table 3.

First, the calibration recovers plausible structural parameter estimates for both regimes. The estimated intervention costs, $\hat{\kappa}_{2.25\%} = 0.0171$ and $\hat{\kappa}_{6\%} = 0.0289$, are roughly two to fifteen times larger than the values used in the numerical exercises of [Miller and Zhang \(1996\)](#), whose unit intervention cost c lies in the range 2×10^{-3} – 8×10^{-3} . This likely reflects the chronic one-sided pressure the lira faced against the depreciation boundary: intervention was not an occasional discrete event, but a persistent commitment requiring repeated policy action. The calibration therefore attributes a substantially higher effective intervention cost to maintaining the target zone. The estimated fundamental volatilities are $\hat{\sigma}_{2.25\%} = 0.0362$ and $\hat{\sigma}_{6\%} = 0.0612$, indicating that the fundamental was considerably more volatile during the wide-band period. This result is consistent with Italy's larger inflation differential and more frequent ERM I realignments during that era.

Second, the counterfactual exercise shows that both structural channels contribute to the

change in the optimal band across regimes, though neither alone is sufficient. Starting from the $\pm 6\%$ baseline trigger $X^* = 6.12\%$, holding $\hat{\kappa}_{6\%}$ fixed and imposing $\hat{\sigma}_{2.25\%}$ yields $X^* = 4.07\%$ (volatility channel, 2.05pp), while holding $\hat{\sigma}_{6\%}$ fixed and imposing $\hat{\kappa}_{2.25\%}$ yields $X^* = 5.47\%$ (cost channel, 0.65pp). The volatility channel therefore accounts for the larger share. Both counterfactual triggers remain well above the $\pm 2.25\%$ band adopted in 1990, and the residual gap cannot be closed by any recombination of the structural parameters; it most plausibly reflects functions the announced band served beyond cost minimisation.

Limitations and extensions Section 2.3 details the assumptions underlying the model and the direction in which each is likely to bias the calibration. Of the four, three bind empirically for the lira – full credibility, zero fundamental drift, and the boundary-symmetric intervention cost – and all three push the estimated $\hat{\kappa}$, and through it the model’s implied optimal band, upward relative to a correctly specified model; the calibrated bands in Table 3 are accordingly best read as upper bounds.

The natural extension is to accommodate stochastic realignment probabilities along the lines of Bertola and Caballero (1992), and to allow κ to differ across the two boundaries. Realignment risk weakens the honeymoon effect, shifts the stationary density away from the boundaries, and introduces an additional cost component for the planner in the form of expected losses from discrete parity jumps; a boundary-specific κ would require separately identifying the frequency and cost of defending each side, rather than pooling them. Taking either extension to data would require joint estimation of within-band dynamics, realignment probabilities, and, for the asymmetric-cost case, boundary-specific intervention records, which we leave for future work.

A Appendix

A.1 Proofs

Proof of Proposition 1. 1. **Exchange rate function.** From the asset pricing equation, the exchange rate is

$$s_t = f(x_t) = \frac{1}{\beta} \left(x_t + \frac{\mathbb{E}[ds_t]}{dt} \right).$$

By Itô's lemma,

$$\frac{\mathbb{E}[ds_t]}{dt} = \frac{\sigma^2}{2} f''(x_t).$$

Hence f solves the ODE

$$f(x) = \frac{1}{\beta} \left(x + \frac{\sigma^2}{2} f''(x) \right),$$

or equivalently,

$$f''(x) - \eta^2 f(x) + \eta^2 \frac{x}{\beta} = 0, \quad \text{where } \eta = \sqrt{\frac{2\beta}{\sigma^2}}.$$

We first solve the homogeneous ODE

$$f''(x) - \eta^2 f(x) = 0,$$

then the solution is

$$f_h(x) = C_1 e^{\eta x} + C_2 e^{-\eta x}.$$

A particular solution is $f_p(x) = x/\beta$, since $f_p''(x) = 0$.

Thus,

$$f(x) = \frac{x}{\beta} + C_1 e^{\eta x} + C_2 e^{-\eta x}.$$

Smooth pasting conditions pin down C_1 and C_2 , but we use symmetric properties of the problem to reduce the problem to finding one constant. Imposing antisymmetry $f(x) = -f(-x)$ gives

$$C_1 = -C_2,$$

so

$$f(x) = \frac{x}{\beta} + C_1 \sinh(\eta x).$$

The smooth-pasting condition requires

$$f'(X) = 0,$$

so

$$0 = \frac{1}{\beta} + C_1 \eta \cosh(\eta X).$$

Solving for C_1 :

$$C_1 = -\frac{1}{\beta\eta \cosh(\eta X)}.$$

Substituting C_1 leads to (5):

$$f(x) = \frac{x}{\beta} - \frac{1}{\beta\eta} \frac{\sinh(\eta x)}{\cosh(\eta X)},$$

which proves the first part.

2. Boundary.

At the boundaries, we impose:

$$f(X) = b, \quad f(-X) = -b.$$

Using antisymmetry, it suffices to use only

$$f(X) = b.$$

Thus,

$$b = \frac{X}{\beta} + C_1 \sinh(\eta X),$$

Substituting C_1 leads to (6):

$$b = \frac{X}{\beta} - \frac{1}{\beta\eta} \tanh(\eta X),$$

which proves the second part.

3. Monotonicity.

Differentiate:

$$\frac{\partial b}{\partial X} = \frac{1}{\beta} \left(1 - \operatorname{sech}^2(\eta X)\right),$$

hence

$$\frac{\partial X}{\partial b} = \frac{\beta}{1 - \operatorname{sech}^2(\eta X)} > 0,$$

since $\tanh'(z) = \operatorname{sech}^2(z) < 1$.

□

Proof of Lemma 1. The stationary density of a driftless diffusion with volatility σ solves the stationary Kolmogorov Forward equation

$$\frac{\sigma^2}{2} m''(x) = 0$$

for $x \in (-X, X)$. General solution is $m(x) = a_0 + a_1 x$. Reflecting barriers require zero flux at each boundary, $m'(-X) = m'(X) = 0$. Since $m'(x) = a_1$ is constant, this forces $a_1 = 0$, so m is constant on $[-X, X]$. Normalizing, $\int_{-X}^X m(x) dx = 2X a_0 = 1$, gives $a_0 = 1/(2X)$. □

Proof of Proposition 2. Using (8) and (9), we substitute $f(x)$:

$$\text{Var}(s) = \frac{1}{2X\beta^2} \int_{-X}^X \left(x - \frac{1}{\eta} \frac{\sinh(\eta x)}{\cosh(\eta X)} \right)^2 dx.$$

Expanding:

$$\text{Var}(s) = \frac{1}{2X\beta^2} (I_1 - I_2 + I_3),$$

where

$$\begin{aligned} I_1 &= \int_{-X}^X x^2 dx, \\ I_2 &= 2 \int_{-X}^X \frac{x}{\eta} \frac{\sinh(\eta x)}{\cosh(\eta X)} dx, \\ I_3 &= \int_{-X}^X \frac{1}{\eta^2} \frac{\sinh^2(\eta x)}{\cosh^2(\eta X)} dx. \end{aligned}$$

$$I_1 = \frac{2X^3}{3}.$$

Using integration by parts,

$$I_2 = \frac{2}{\eta \cosh(\eta X)} \int_{-X}^X x \sinh(\eta x) dx = \frac{4X}{\eta^2} - \frac{4 \tanh(\eta X)}{\eta^3}.$$

Using $\sinh^2(z) = \frac{\cosh(2z)-1}{2}$,

$$I_3 = \frac{1}{\eta^2 \cosh^2(\eta X)} \int_{-X}^X \sinh^2(\eta x) dx = -\frac{X}{\eta^2 \cosh^2(\eta X)} + \frac{\tanh(\eta X)}{\eta^3}.$$

Substituting I_1 , I_2 , and I_3 :

$$\text{Var}(s) = \frac{1}{2X\beta^2} \left(\frac{2X^3}{3} - \frac{4X}{\eta^2} + \frac{5 \tanh(\eta X)}{\eta^3} - \frac{X}{\eta^2 \cosh^2(\eta X)} \right),$$

which proves (10). □

Proof of Corollary 1. 1. **Small X .**

Taking $\eta X \rightarrow 0$. Since $|x| \leq X$, it follows that $|\eta x| \leq \eta X$, so that $\eta x \rightarrow 0 \quad \forall x \in [-X, X]$.

Using Taylor expansions around zero

$$\sinh(\eta x) \approx \eta x + \frac{(\eta x)^3}{6}, \quad \frac{1}{\cosh(\eta X)} \approx 1 - \frac{(\eta X)^2}{2}.$$

Substituting into $f(x)$,

$$f(x) \approx \frac{1}{\beta} \left[x - \frac{1}{\eta} \left(\eta x + \frac{\eta^3 x^3}{6} \right) \left(1 - \frac{\eta^2 X^2}{2} \right) \right].$$

Expanding and keeping leading terms

$$f(x) \approx \frac{1}{\beta} \left[x - \left(x + \frac{\eta^2 x^3}{6} - \frac{\eta^2 X^2 x}{2} \right) \right].$$

Thus,

$$f(x) \approx \frac{1}{\beta} \left(\frac{\eta^2 X^2}{2} x - \frac{\eta^2}{6} x^3 \right).$$

Under symmetry and ergodicity, the stationary density is uniform:

$$m(x; X) = \frac{1}{2X}, \quad x \in [-X, X].$$

Hence,

$$\text{Var}(s) = \frac{1}{2X} \int_{-X}^X f(x)^2 dx.$$

Using symmetry:

$$\text{Var}(s) = \frac{1}{X} \int_0^X f(x)^2 dx.$$

Substitute $f(x)$:

$$f(x)^2 = \frac{\eta^4}{\beta^2} \left(\frac{X^2}{2} x - \frac{1}{6} x^3 \right)^2.$$

Compute:

$$\text{Var}(s) = \frac{\eta^4}{\beta^2 X} \int_0^X \left(\frac{X^2}{2} x - \frac{1}{6} x^3 \right)^2 dx.$$

Evaluating the integral gives:

$$\text{Var}(s) \approx \frac{17}{315} \frac{\eta^4 X^6}{\beta^2}.$$

To derive the expression for b we use a third-order Taylor approximation of (6) when X is small. So using

$$\tanh(\eta X) \approx \eta X - \frac{(\eta X)^3}{3},$$

we get

$$b \approx \frac{2X^3}{3\sigma^2},$$

which after the substitution into the variance leads to

$$\text{Var}(s) \approx \frac{17}{35} b^2.$$

2. Large X .

As $\eta X \rightarrow \infty$, $\sinh(\eta x)$ grows no faster than $\frac{1}{2}e^{\eta|x|}$,

while

$$\cosh(\eta X) \rightarrow \infty.$$

Thus,

$$\frac{\sinh(\eta x)}{\cosh(\eta X)} \rightarrow 0 \quad \text{for } |x| < X.$$

Hence,

$$f(x) \approx \frac{x}{\beta} \quad \text{and} \quad b \approx X/\beta.$$

Therefore,

$$\text{Var}(s) \approx \frac{1}{\beta^2} \text{Var}(x).$$

Since $x \sim \mathcal{U}[-X, X]$:

$$\text{Var}(x) = \frac{X^2}{3}.$$

Thus,

$$\text{Var}(s) \approx \frac{X^2}{3\beta^2}.$$

Substituting $b \approx X/\beta$ into the variance we obtain the equivalent expression under the given approximation

$$\text{Var}(s) \approx \frac{1}{3} b^2.$$

□

Proof of Lemma 2. Let φ be any twice continuously differentiable test function. Applying Itô's formula to $\varphi(x_t)$ and using that dU_t is supported on $\{x_t = -X\}$ and dD_t on $\{x_t = X\}$,

$$d\varphi(x_t) = \sigma\varphi'(x_t) dW_t + \varphi'(-X) dU_t - \varphi'(X) dD_t + \frac{\sigma^2}{2} \varphi''(x_t) dt.$$

Taking expectations, dividing by t , and letting $t \rightarrow \infty$, the martingale term vanishes, $\mathbb{E}[\varphi(x_t)]/t \rightarrow 0$ since $\varphi(x_t)$ is bounded on $[-X, X]$, and the ergodic theorem gives

$$\frac{1}{t} \mathbb{E} \int_0^t \varphi''(x_s) ds \rightarrow \int_{-X}^X \varphi''(x) m(x; X) dx.$$

Hence

$$0 = \varphi'(-X) A_u(X) - \varphi'(X) A_d(X) + \frac{\sigma^2}{2} \int_{-X}^X \varphi''(x) m(x; X) dx.$$

Taking $\varphi(x) = x$, so $\varphi' \equiv 1$ and $\varphi'' \equiv 0$, then $A_u(X) = A_d(X)$, recovering the symmetry claim directly from the dynamics rather than by appeal to symmetry alone. Taking $\varphi(x) = x^2/2$, so $\varphi'(x) = x$ and $\varphi'' \equiv 1$, and using $m(x; X) = 1/(2X)$ with $\int_{-X}^X m(x; X) dx = 1$,

$$0 = -X A_u(X) - X A_d(X) + \frac{\sigma^2}{2}.$$

Substituting $A_u(X) = A_d(X)$ and solving gives $A_u(X) = A_d(X) = \sigma^2/(4X)$.

□

Proof of Proposition 3. 1. **Exact solution.** Differentiating the exact objective (15) with respect to X and setting the derivative to zero gives

$$\frac{2X}{3\beta^2} - \frac{5 \tanh(\eta X)}{2\eta^3 X^2 \beta^2} + \frac{5}{2\eta^2 X \beta^2 \cosh^2(\eta X)} + \frac{\sinh(2\eta X)}{2\eta \beta^2 \cosh^4(\eta X)} - \frac{\kappa \sigma^2}{2X^2} = 0.$$

To establish uniqueness the second derivative should be taken:

$$\begin{aligned} \frac{2}{3\beta^2} - \frac{5 \operatorname{sech}^2(\eta X)}{\eta^2 \beta^2 X^2} + \frac{5 \tanh(\eta X)}{\eta^3 \beta^2 X^3} - \frac{5 \tanh(\eta X) \operatorname{sech}^2(\eta X)}{\eta \beta^2 X} + \\ + \frac{\operatorname{sech}^2(\eta X) (\operatorname{sech}^2(\eta X) - 2 \tanh^2(\eta X))}{\beta^2} + \frac{\kappa \sigma^2}{X^3}, \end{aligned}$$

then multiplying both sides of F.O.C. by $2/X$ and substituting $\frac{\kappa \sigma^2}{X^3}$, this expression simplifies to

$$\frac{1}{\beta^2} \left[2 - \operatorname{sech}^2(\eta X) \left(\frac{3 \tanh(\eta X)}{\eta X} - \operatorname{sech}^2(\eta X) + 2 \tanh^2(\eta X) \right) \right].$$

Defining $t := \eta X$ and $\psi(t) := 2 - \operatorname{sech}^2(t) \left(\frac{3 \tanh(t)}{t} - \operatorname{sech}^2(t) + 2 \tanh^2(t) \right)$ the problem reduces to determining the sign of $\psi(t)$ for $t > 0$. If it positive, then S.O.C. for the unique minimum is satisfied.

2. **Narrow-band approximation.** Using the approximation (11), we get the following problem

$$\min_X \frac{17}{315} \frac{\eta^4 X^6}{\beta^2} + \frac{\kappa \sigma^2}{2X},$$

F.O.C. gives

$$\frac{34}{105} \frac{\eta^4 X^5}{\beta^2} - \frac{\kappa \sigma^2}{2X^2} = 0.$$

The first derivative is strictly convex function for $X > 0$ as the second derivative is strictly positive for $X > 0$:

$$\frac{34\eta^4 X^4}{21\beta^2} + \frac{\kappa \sigma^2}{X^3} > 0,$$

hence the F.O.C. gives unique interior minimizer. Solving for optimal band, one can obtain

$$X = \left(\frac{105\kappa\sigma^6}{272} \right)^{1/7}.$$

To derive the expression for b we use the following approximation for b :

$$b \approx \frac{2X^3}{3\sigma^2}.$$

Finally,

$$b = \frac{2}{3} \left(\frac{105}{272} \right)^{3/7} \sigma^{4/7} \kappa^{3/7}.$$

3. **Wide-band approximation.** The approximated expression (12) for the boundaries when ηX is large, the corresponding problem becomes

$$\min_X \frac{X^2}{3\beta^2} + \frac{\kappa\sigma^2}{2X}$$

F.O.C. gives

$$\frac{2X}{3\beta^2} - \frac{\kappa\sigma^2}{2X^2} = 0.$$

The first derivative strictly convex for $X > 0$ as the second derivative is strictly positive for $X > 0$:

$$\frac{2}{3\beta^2} + \frac{\kappa\sigma^2}{X^3} > 0,$$

hence the F.O.C. gives a unique interior minimizer.

The optimal band will be

$$X = \left(\frac{3\kappa\sigma^2\beta^2}{4} \right)^{1/3},$$

using that for large boundaries $b \approx X/\beta$, we get the final expression

$$b = \left(\frac{3\kappa\sigma^2}{4\beta} \right)^{1/3}.$$

□

Proof of Corollary 2. Set $v := X/\sigma$, so $X = \sigma v$. Since $\eta = \sqrt{2\beta}/\sigma$, we have $\eta X = \sqrt{2\beta} v$, and the F.O.C. (16) becomes, after substituting $X = \sigma v$ and $\eta = \sqrt{2\beta}/\sigma$,

$$\frac{2\sigma v}{3\beta^2} - \frac{5\sigma \tanh(\sqrt{2\beta} v)}{4\sqrt{2}\beta^{7/2}v^2} + \frac{5\sigma}{4\beta^3 v \cosh^2(\sqrt{2\beta} v)} + \frac{\sqrt{2}\sigma \sinh(2\sqrt{2\beta} v)}{2\beta^{5/2} \cosh^4(\sqrt{2\beta} v)} - \frac{\kappa}{2v^2} = 0.$$

Multiplying through by $2v^2/\sigma > 0$ yields

$$g(v) = \frac{\kappa}{\sigma},$$

where

$$g(v) = \frac{4v^3}{3\beta^2} - \frac{5 \tanh(\sqrt{2\beta} v)}{2\sqrt{2}\beta^{7/2}} + \frac{5v}{2\beta^3 \cosh^2(\sqrt{2\beta} v)} + \frac{\sqrt{2}v^2 \sinh(2\sqrt{2\beta} v)}{2\beta^{5/2} \cosh^4(\sqrt{2\beta} v)}.$$

Taking the derivative w.r.t. v one could notice that $g'(v) > 0$:

$$g'(v) = \frac{1}{\beta^3} \left[3 \left(\sqrt{2\beta} v \right)^2 - 4 \left(\sqrt{2\beta} v \right)^2 \tanh^2 \left(\sqrt{2\beta} v \right) + 3 \left(\sqrt{2\beta} v \right)^2 \tanh^4 \left(\sqrt{2\beta} v \right) - \right. \\ \left. - 3 \left(\sqrt{2\beta} v \right) \tanh \left(\sqrt{2\beta} v \right) \left(1 - \tanh^2 \left(\sqrt{2\beta} v \right) \right) \right] > 0.$$

Since g is strictly increasing on $(0, +\infty)$, the equation $g(v^*) = \kappa/\sigma$ has a unique solution $v^* = g^{-1}(\kappa/\sigma)$, which depends only on the ratio κ/σ . Reverting to the original variable gives $X^* = \sigma \phi(\kappa/\sigma)$, establishing homogeneity of degree 1 in (κ, σ) .

Parts 2 and 3 are immediate: dividing the approximations (17)–(18) by σ gives

$$v^* = \frac{X^*}{\sigma} = \begin{cases} \left(\frac{105}{272}\right)^{1/7} \left(\frac{\kappa}{\sigma}\right)^{1/7} & \text{(narrow-band),} \\ \left(\frac{3\beta^2}{4}\right)^{1/3} \left(\frac{\kappa}{\sigma}\right)^{1/3} & \text{(wide-band),} \end{cases}$$

which are exactly $\phi(\kappa/\sigma)$ in parts (2) and (3). □

A.2 Calibration algorithms

This section describes the main algorithms used in the calibration.

Main algorithm The algorithm recovers the three structural parameters $(\hat{\sigma}, \hat{X}, \hat{\kappa})$ from two observable moments – the observed band half-width b_{obs} and the sample variance $\widehat{\text{Var}}(s)$ – and the known parameter β . The key step exploits the homogeneity property (Corollary 2): dividing the variance equation by the square of the boundary equation eliminates σ and yields a single equation in $v = X/\sigma$ alone, which is solved numerically. The remaining parameters follow by back-substitution.

Algorithm 1 Parameters Calibration

Input: $\widehat{\text{Var}}(s)$: empirical variance of the exchange rate within the regime

b_{obs} : observed band half-width (0.06 or 0.0225)

β : money demand semi-elasticity parameter

Output: $\hat{\sigma}$: fundamental volatility, $\hat{\kappa}$: intervention cost

Solve for $\hat{v}$ via the boundary–variance moment condition

$$b_{\text{obs}}^2 \cdot q(\hat{v}, \beta) - \widehat{\text{Var}}(s) \cdot h(\hat{v}, \beta)^2 = 0,$$

using Newton’s method in log space $x = \log v$, with initial guess

$$v_0 = \frac{b_{\text{obs}}}{\sqrt{\widehat{\text{Var}}(s)}}.$$

Recover fundamental volatility from the boundary equation:

$$\hat{\sigma} = \frac{b_{\text{obs}}}{h(\hat{v}, \beta)}.$$

Recover the normalized cost $\hat{\theta} = \hat{\kappa}/\hat{\sigma}$ from the first-order condition:

$$\hat{\theta} = g(\hat{v}, \beta).$$

Recover the intervention cost:

$$\hat{\kappa} = \hat{\theta} \cdot \hat{\sigma}.$$

Newton algorithm This algorithm uses reparameterization to avoid negative iterates and improve scaling. In principle, it works in log space: $x = \log v$, so $v = e^x > 0$ always.

Algorithm 2 Newton's Method in Log Space for $f(v) = 0$

Input: $f(v)$, $f'(v)$: objective and its derivative

v_0 : initial guess for v

ε : convergence tolerance

N : maximum number of iterations

Output: $\hat{v}$: root of f

Set $x_0 \leftarrow \log v_0$

for $k = 1, 2, \dots, N$ **do**

$v \leftarrow e^{x_{k-1}}$

Evaluate objective and derivative at current v :

$$f_k \leftarrow f(v), \quad f'_k \leftarrow f'(v)$$

Apply chain rule to convert to x -space derivative:

$$\left. \frac{d}{dx} f(e^x) \right|_{x_{k-1}} = f'(v) \cdot v = f'_k \cdot v$$

if $|f_k/(f'_k \cdot v)| > C$ **then**

abort

▷ derivative too small relative to f , step would explode

end if

Newton step in log space:

$$x_k \leftarrow x_{k-1} - \frac{f_k}{f'_k \cdot v}$$

if $|f_k| < \varepsilon$ **then**

return $\hat{v} \leftarrow e^{x_k}$

▷ converged

end if

end for

error: did not converge within N iterations

A.3 Plots

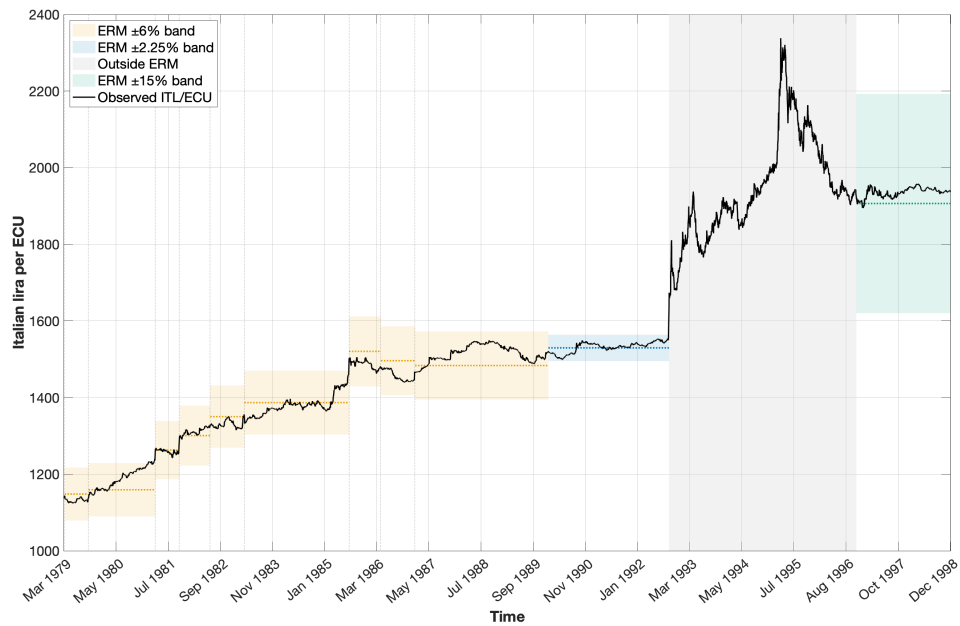


Figure 6: Italian lira per ECU over the full ERM I history (March 1979 – December 1998), across the $\pm 6\%$, $\pm 2.25\%$, outside-ERM I, and $\pm 15\%$ regimes.

Source: Based on ECB daily data.

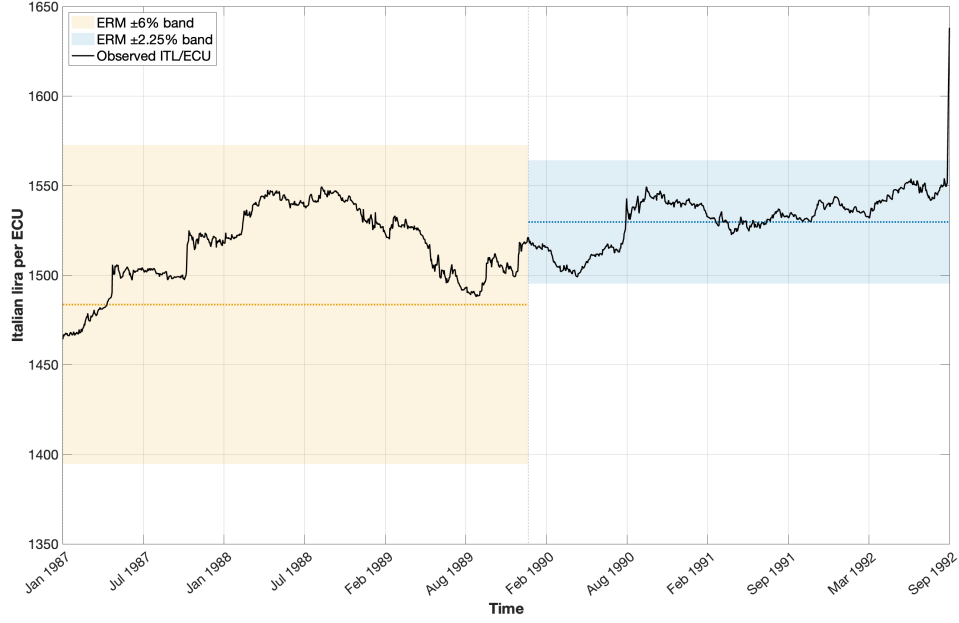


Figure 7: Italian lira per ECU over the 1987–1992 estimation window (12 January 1987 – 16 September 1992), spanning the transition from the $\pm 6\%$ to the $\pm 2.25\%$ ERM I band in January 1990.

Source: Based on ECB daily data.

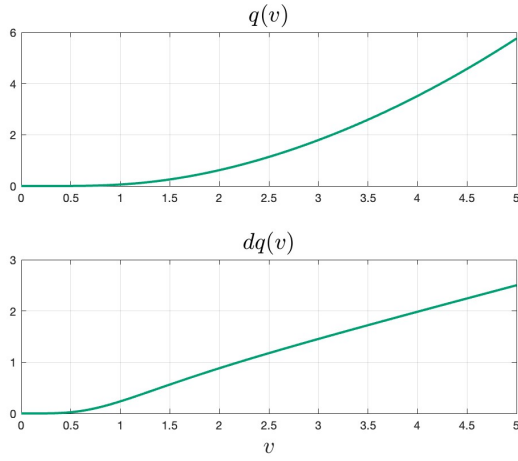


Figure 8: Function $q(v)$ and $q'(v)$.

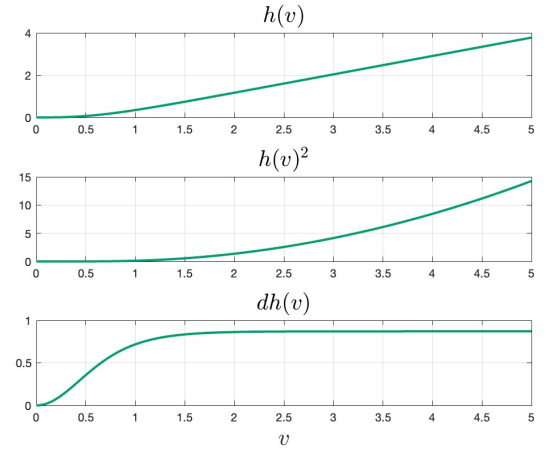


Figure 9: Functions $h(v)$, $h^2(v)$ and $h'(v)$.

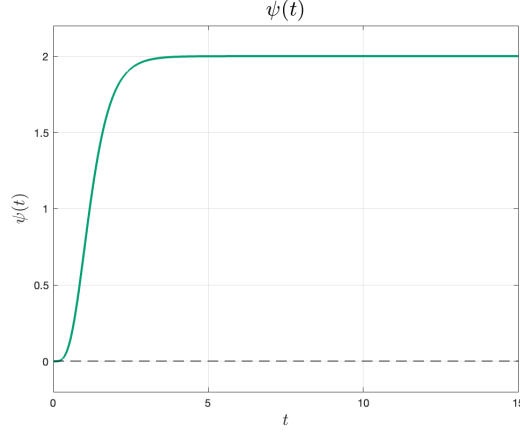


Figure 10: Function $\psi(t)$ is positive ($\in (0, 2)$ for $\forall t \in (0, +\infty)$) on the positive domain.

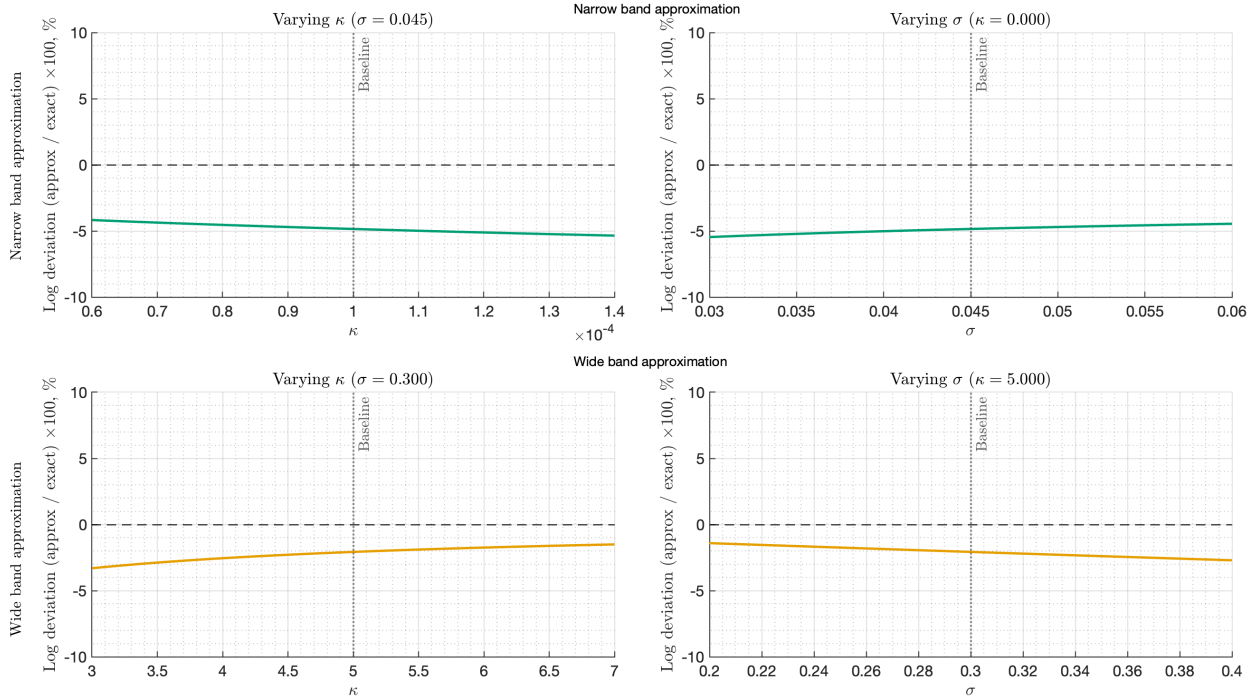


Figure 11: Accuracy of the optimal-band approximations as the structural parameters vary one at a time. Each panel plots the log deviation of the approximate optimal band from the exact solution in percentage, $100 \times \log(X_{\text{approx}}/X_{\text{exact}})$; the horizontal dashed line marks zero error and the dotted vertical line the reference value. *Top row*: narrow-band approximation, varying κ (at $\sigma = 0.045$) and σ (at $\kappa = 0.010$). *Bottom row*: wide-band approximation, varying κ (at $\sigma = 0.300$) and σ (at $\kappa = 0.020$). Each approximation crosses zero near its reference calibration, so the error is small in the relevant neighbourhood and grows as parameters move away from it.

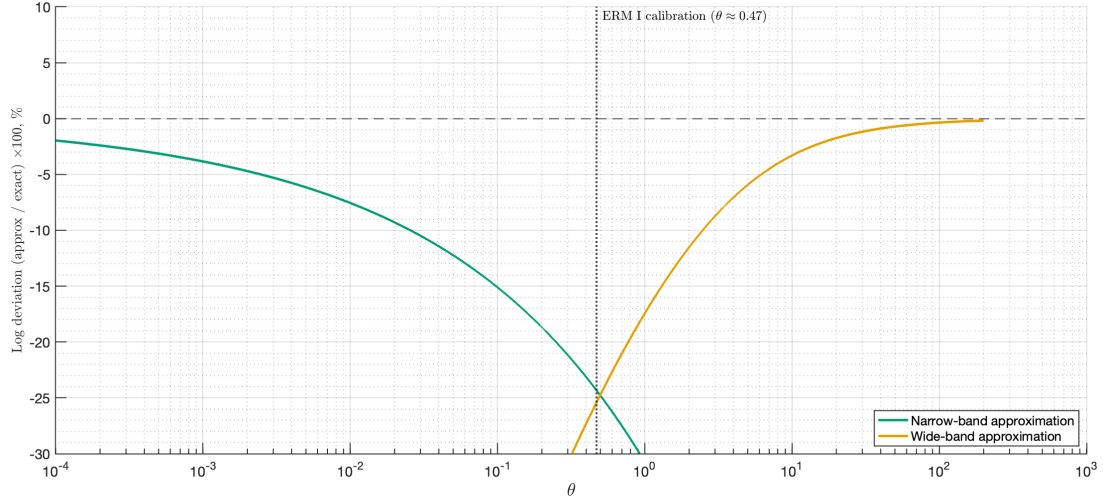


Figure 12: Relative error of the two band approximations as a function of $\theta := \kappa/\sigma$ (log scale), measured as $100 \times \log(X_{approx}/X_{exact})$. The vertical dashed line marks the calibrated $\theta \approx 0.47$ common to both ERM I regimes, equivalent to $\eta X \approx 1.52$ – an intermediate value in neither the narrow- nor the wide-band asymptotic regime.

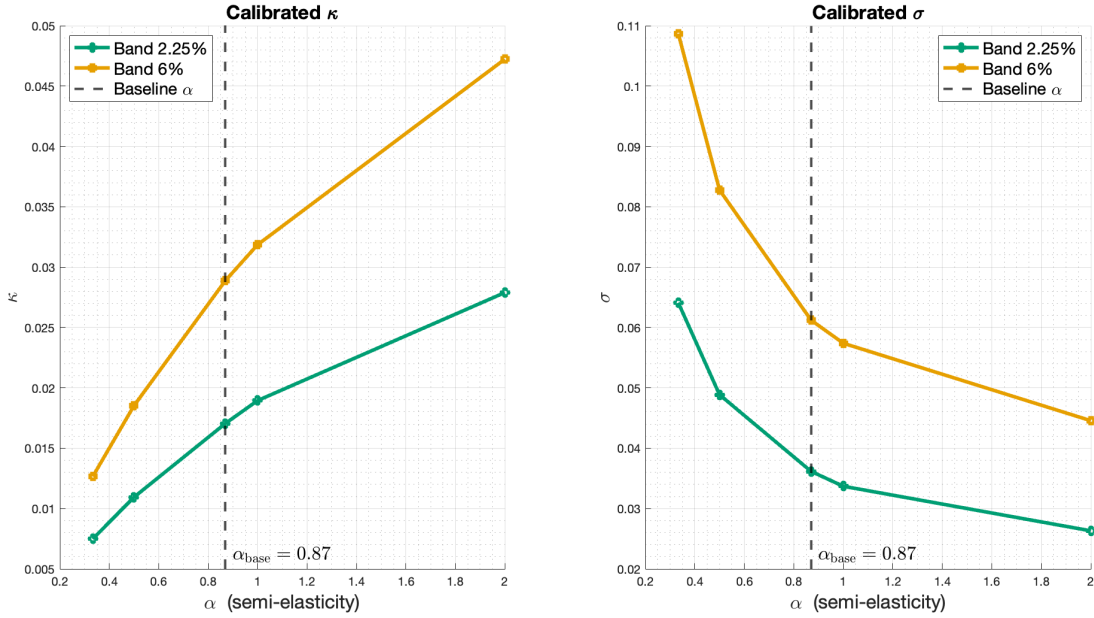


Figure 13: Robustness of the calibrated structural parameters to the assumed money-demand semi-elasticity α (equivalently $\beta = 1/\alpha$). For each α the inverse problem is re-solved on each regime, recovering the intervention cost κ (left) and the fundamental volatility σ (right); the dashed vertical line marks the baseline $\alpha_{base} = 0.87$.

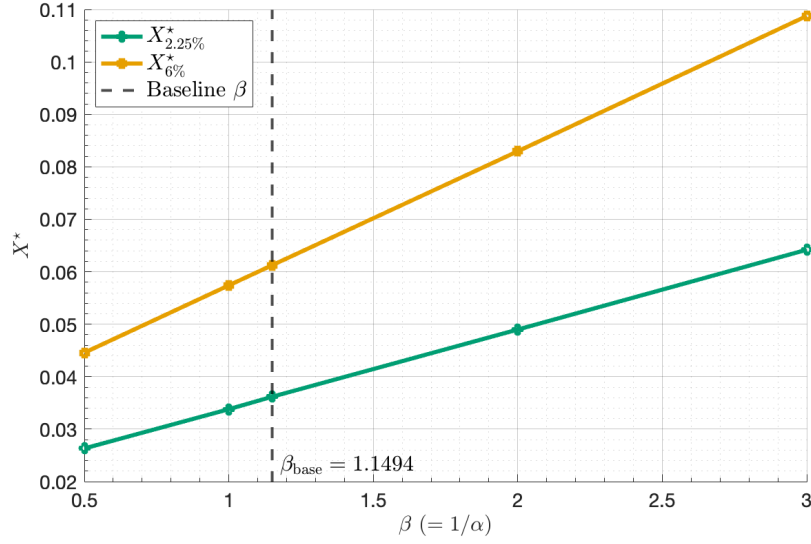


Figure 14: Robustness of the optimal intervention trigger to the money-demand semi-elasticity. X^* is the optimal fundamental boundary in each regime, plotted against $\beta = 1/\alpha$. The dashed vertical line marks the baseline $\beta_{\text{base}} = 1.1494$. The optimal trigger rises roughly linearly in β in both regimes, and the $\pm 6\%$ regime calls for a wider trigger than the $\pm 2.25\%$ regime at every β , so the ranking of the two regimes does not depend on the choice of β .

References

- Alvarez, Fernando.** 2025. “Widening currency bands.” Einaudi Institute for Economics and Finance.
- Bertola, Giuseppe, and Ricardo J. Caballero.** 1992. “Target Zones and Realignment.” *The American Economic Review*, 82(3): 520–536.
- Coenen, G., and J.-L. Vega.** 2001. “The Demand for M3 in the Euro Area.” *Journal of Applied Econometrics*, 16(6): 727–748.
- Constantinides, George M., and Scott F. Richard.** 1978. “Existence of Optimal Simple Policies for Discounted-Cost Inventory and Cash Management in Continuous Time.” *Operations Research*, 26(4): 620–636.
- Detragiache, Enrica, and A. Javier Hamann.** 1999. “Exchange Rate-Based Stabilization In Western Europe: Greece, Ireland, Italy, And Portugal.” *Contemporary Economic Policy*, 17(3): 358–369.
- Im, Joon-Hwan.** 2001. “Optimal Currency Target Zones: How Wide Should Exchange Rate Bands Be?” *International Economic Journal*, 15(1): 61–93.
- Krugman, Paul.** 1979. “A Model of Balance-of-Payments Crises.” *Journal of Money, Credit and Banking*, 11(3): 311–325. Publisher: [Wiley, Ohio State University Press].
- Krugman, Paul.** 1991. “Target Zones and Exchange Rate Dynamics.” *The Quarterly Journal of Economics*, 106(3): 669–682. Publisher: Oxford University Press.
- Miller, Marcus, and Lei Zhang.** 1996. “Optimal target zones: How an exchange rate mechanism can improve upon discretion.” *Journal of Economic Dynamics and Control*, 20(9): 1641–1660.
- Miller, Merton H., and Daniel Orr.** 1966. “A Model of the Demand for Money by Firms.” *The Quarterly Journal of Economics*, 80(3): 413–435. Publisher: Oxford University Press.
- Svensson, Lars E.O.** 1991. “Target zones and interest rate variability.” *Journal of International Economics*, 31(1): 27–54.
- Svensson, Lars E. O.** 1992. “An Interpretation of Recent Research on Exchange Rate Target Zones.” *Journal of Economic Perspectives*, 6(4): 119–144.
- Svensson, Lars E.O.** 1994. “Why exchange rate bands?: Monetary independence in spite of fixed exchange rates.” *Journal of Monetary Economics*, 33(1): 157–199.